

Grade 12
Pre-Calculus Mathematics
Achievement Test

Marking Guide

January 2025

Grade 12 Pre-Calculus Mathematics Achievement Test:
Marking Guide (January 2025)

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While the department is committed to making its publications as accessible as possible, some parts of this document are not fully accessible at this time.

Available in alternate formats upon request.

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General Marking Instructions

Please do not make any marks in the student test booklets. If the booklets have marks in them, the marks will need to be removed by departmental staff prior to sample marking should the booklet be selected.

Please ensure that

- the booklet number and the number on the *Answer/Scoring Sheet* are identical
- **students and markers use only a pencil to complete the *Answer/Scoring Sheets***
- the totals of each of the four parts are written at the bottom
- each student's final result is recorded, by booklet number, on the corresponding *Answer/Scoring Sheet*
- the *Answer/Scoring Sheet* is complete
- a photocopy has been made for school records

Once marking is completed, please forward the *Answer/Scoring Sheets* to Manitoba Education and Early Childhood Learning in the envelope provided. For more information see the administration manual.

Marking the Test Questions

The test is composed of constructed response questions and selected response questions. Constructed response questions are worth 1 to 5 marks each, and selected response questions are worth 1 mark each. An answer key for the selected response questions can be found at the beginning of the section "Booklet 2 Questions."

To receive full marks, a student's response must be complete and correct. Where alternative answering methods are possible, the *Marking Guide* attempts to address the most common solutions. For general guidelines regarding the scoring of students' responses, see Appendix A.

Irregularities in Provincial Tests

During the administration of provincial tests, supervising teachers may encounter irregularities. Markers may also encounter irregularities during local marking sessions. The appendix provides examples of such irregularities as well as procedures to follow to report irregularities.

If a *Answer/Scoring Sheet* is marked with "0" only (e.g., student was present but did not attempt any questions) please document this on the *Irregular Test Booklet Report*.

Assistance

If, during marking, any marking issue arises that cannot be resolved locally, please call Manitoba Education and Early Childhood Learning at the earliest opportunity to advise us of the situation and seek assistance if necessary.

You must contact the Assessment Consultant responsible for this project before making any modifications to the answer keys or scoring rubrics.

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Communication Errors

The marks allocated to questions are primarily based on the concepts and procedures associated with the learning outcomes in the curriculum. For each question, shade in the circle on the *Answer/Scoring Sheet* that represents the marks given based on the concepts and procedures. A total of these marks will provide the preliminary mark.

Errors that are not related to concepts or procedures are called “Communication Errors” (see Appendix A) and will be tracked on the *Answer/Scoring Sheet* in a separate section. There is a $\frac{1}{2}$ mark deduction for each type of communication error committed, regardless of the number of errors per type (i.e., committing a second error for any type will not further affect a student’s mark), with a maximum deduction of 5 marks from the total test mark.

When a given response includes multiple types of communication errors, deductions are indicated in the order in which the errors occur in the response. No communication errors are recorded for work that has not been awarded marks. The total deduction may not exceed the marks awarded.

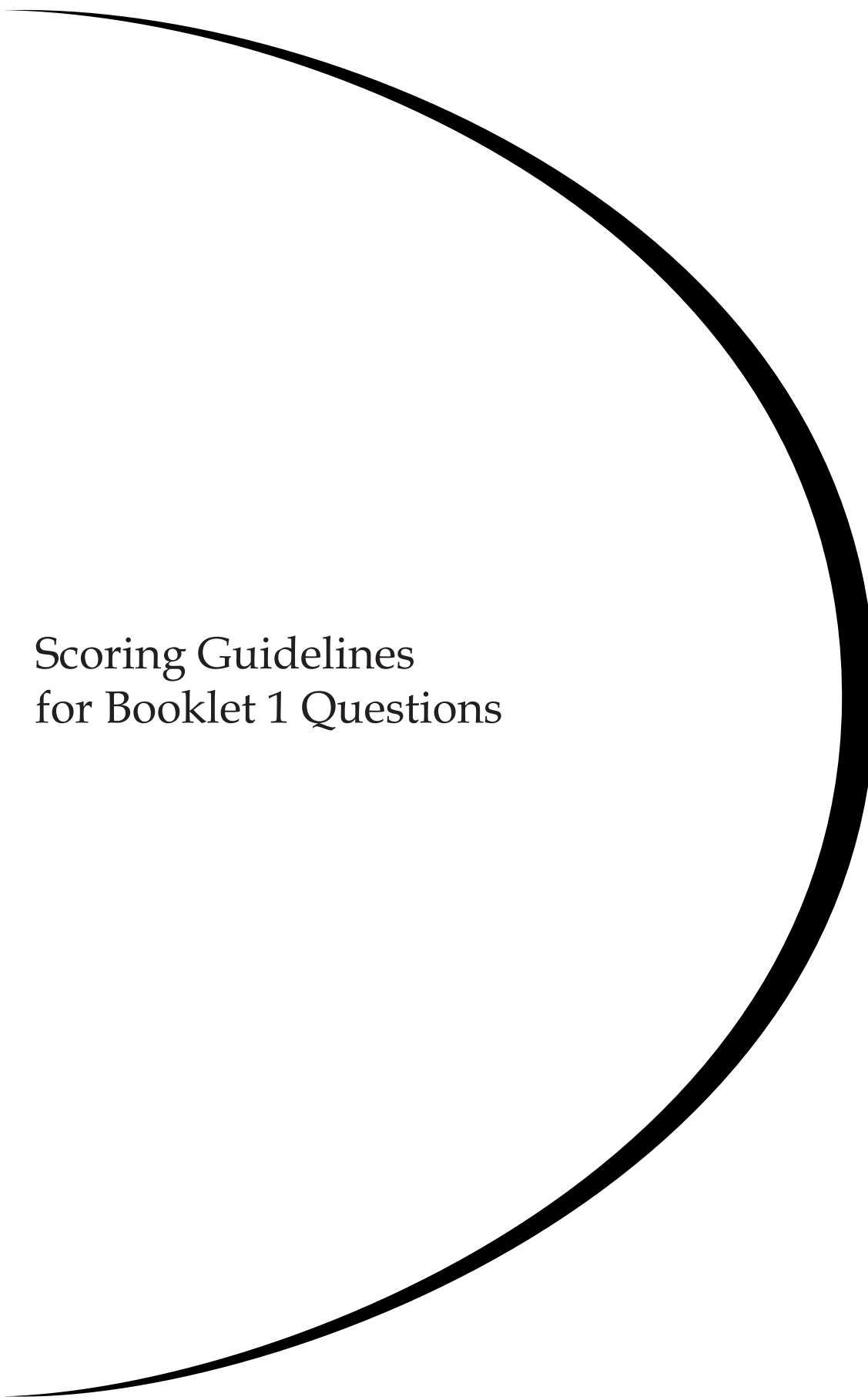
The student’s final mark is determined by subtracting the communication errors from the preliminary mark.

Example: A student has a preliminary mark of 72. The student committed two E1 errors ($\frac{1}{2}$ mark deduction), four E7 errors ($\frac{1}{2}$ mark deduction), and one E8 error ($\frac{1}{2}$ mark deduction). Although seven communication errors were committed in total, there is a deduction of only $1\frac{1}{2}$ marks.

COMMUNICATION ERRORS / ERREURS DE COMMUNICATION									
Shade in the circles below for a maximum total deduction of 5 marks ($\frac{1}{2}$ mark deduction per error). Noircir les cercles ci-dessous pour une déduction maximale totale de 5 points (déduction de 0,5 point par erreur).									
E1	☒	E2	☐	E3	☐	E4	☐	E5	☐
E6	☐	E7	☒	E8	☒	E9	☐	E10	☐

Example: Marks assigned to the student

Marks Awarded	Booklet 1	Selected Response	Booklet 2	Communication Errors (Deduct)	Total
	25	7	40	$1\frac{1}{2}$	$70\frac{1}{2}$
Total Marks	36	9	45	Maximum deduction of 5 marks	90



Scoring Guidelines for Booklet 1 Questions

Question 1

P1

Youyi wants to buy a new cell phone from a company that offers 33 different types of cell phones. Each cell phone is available in 5 different colours. Determine the total number of available options.

Solution

$$\underline{33} \bullet \underline{5} = 165 \text{ options}$$

1 mark

Exemplar 1

33 type of telephones
5 Colours

$${}^{33}C_5$$

$$\frac{33!}{(33-5)!5!} \\ = 237336 \text{ available options}$$

0 out of 1

Exemplar 2

165

1 out of 1

Exemplar 3

$${}^{33}P_5 = 28480320 \text{ possibilities}$$

0 out of 1

Question 2

T1

Determine the central angle subtended by an arc length of 30 cm on a circle with a diameter of 20.6 cm. Express your answer in degrees.

Solution

$$s = \theta r$$

1 mark for substitution

$$30 = \theta(10.3)$$

$$\theta = 2.912\ 621\dots$$

$$\theta = (2.912\ 621)\left(\frac{180^\circ}{\pi}\right)$$

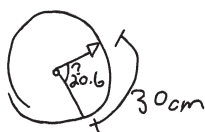
1 mark for conversion

$$\theta = 166.880\ 911\dots^\circ$$

$$\theta = 166.881^\circ$$

2 marks

Exemplar 1



$$\begin{aligned} s &= \theta r \\ 30 &= \theta 20.6 \\ \frac{30}{20.6} & \\ \theta &= 1.45631 \text{ rad} \\ &\times \frac{180}{\pi} \\ \theta &= 83.4398^\circ \end{aligned}$$

1½ out of 2

award full marks

– ½ mark for procedural error (substituting diameter instead of radius)

E6 (rounding too early)

E7 (notation error in line 2)

Exemplar 2

$$\frac{s}{r} = \frac{\theta r}{r}$$

$$\begin{aligned} R &= \frac{20.6}{2} \\ R &= 10.3 \end{aligned}$$

$$\theta = \frac{s}{r}$$

$$\theta = \frac{30}{20.6} \times \frac{180}{\pi}$$

$$\theta = 83.440^\circ$$

2 out of 2

award full marks

E7 (transcription error in line 3)

Exemplar 3

$$s = 30$$

$$r = \frac{20.6 \text{ cm}}{2}$$

$$r = 10.3$$

$$s = \theta r$$

$$\frac{30}{10.3} = \frac{\theta (10.3)}{10.3}$$

$$\theta = 2.913$$

1 out of 2

+ 1 mark for substitution

Question 3

P3

A choir has 14 sopranos. Karin and Ellen are two of them. Determine the number of possible ensembles of 3 sopranos that could be formed if either Karin or Ellen, or both, must be included.

Solution

Method 1

$$\text{Case 1: } {}_1C_1 \cdot {}_{12}C_2 = 66$$

1 mark for Case 1 and Case 2

$$\text{Case 2: } {}_1C_1 \cdot {}_{12}C_2 = 66$$

$$\text{Case 3: } {}_2C_2 \cdot {}_{12}C_1 = 12$$

1 mark for Case 3

$$66 + 66 + 12 = 144$$

1 mark for addition of cases

3 marks

Method 2

$${}_{14}C_3 - {}_{12}C_3 = 144$$

1 mark for ${}_{14}C_3$

1 mark for ${}_{12}C_3$

1 mark for subtraction of cases

3 marks

Exemplar 1

Karin Ellen 12 = 12 ensembles can be created.

1 out of 3

+1 mark for case 3

Exemplar 2

$$\text{Karin: } \underline{1} \times \underline{13} \times \underline{12} = 156$$

$$\text{Ellen: } \underline{1} \times \underline{13} \times \underline{12} = 156$$

$$\text{Both: } \underline{1} \times \underline{1} \times \underline{12} = \underline{12}$$

324 ensembles

2 out of 3

+ 1 mark for case 3

+ 1 mark for addition of cases

Exemplar 3

$$\text{case 1 (Karin): } = {}_1C_1 \cdot {}_{13}C_2 = 78$$

$$\text{Case 2 (Ellen): } {}_1C_1 \cdot {}_{13}C_2 = 78$$

$$\text{Case 3 (both): } {}_2C_2 \cdot {}_{12}C_1 = 12$$

$$168$$

2 out of 3

+ 1 mark for case 3

+ 1 mark for addition of cases

Question 4

R10

Loïc invests \$5000 at an annual interest rate of 6.75% compounded semi-annually. The value of his investment grows according to the following formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A is the value of the investment after t years

P is the amount of the initial deposit

r is the annual interest rate (as a decimal)

n is the number of compounding periods per year

t is the time in years

Determine, algebraically, the length of time it will take Loïc's investment to grow to \$9500.

Solution

$$9500 = 5000 \left(1 + \frac{0.0675}{2} \right)^{2t} \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.9 = (1.03375)^{2t}$$

$$\log 1.9 = \log(1.03375)$$

1 mark for applying logarithms

$$\log 1.9 = 2t \log 1.03375$$

1 mark for power law (or appropriate logarithm laws)

$$t = \frac{\log 1.9}{2 \log 1.03375}$$

$$t = 9.668\ 522\dots$$

$\frac{1}{2}$ mark for evaluating a quotient of logarithms

$$t = 9.669 \text{ years}$$

$$\therefore t = 10 \text{ years}$$

3 marks

Note:

Accept $t = 9.669$ years as a final answer (or consistent value of t).

Exemplar 1

$$A = 9500$$

$$P = 5000$$

$$r = 0.0675$$

$$n = 12$$

$$t = ?$$

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

$$\frac{9500}{5000} = \frac{5000}{5000} \left(1 + \frac{0.0675}{12} \right)^{12t}$$

$$1.9 = \left(1 + 0.005625 \right)^{12t}$$

$$1.9 = 1.005625^{12t}$$

$$\log 1.9 = \log 1.005625^{12t}$$

$$0.278753601 = 12t \log 1.005625$$

$$\frac{0.278753601}{0.02923...} = \frac{0.02923273732t}{0.02923...}$$

$$9.535665373 = t$$

$$\boxed{10 \text{ years}}$$

2½ out of 3

+ 1 mark for applying logarithms

+ 1 mark for power law

+ ½ mark for evaluating a quotient of logarithms

Exemplar 2

$$9500 = 5000 \left(1 + \frac{0.0675}{12}\right)^{12t}$$

$$\frac{9500}{5000} = (1.005625)^{12t}$$

$$\log 1.8 = (12t) \log(1.005625)$$

$$\frac{\log 1.8}{\log 1.005625} = 12t$$

$$\frac{104.7890257}{12} = \frac{12t}{12}$$

$$8.732 = t$$

It will take Loic 8.732 years.

2 out of 3

+ 1 mark for applying logarithms

+ 1 mark for power law

+ ½ mark for evaluating a quotient of logarithms

– ½ mark for arithmetic error in line 3

Exemplar 3

invests \$5000
annual interest 6.75 % 2 x per year

$$9500 = 5000 \left(1 + \frac{0.0675}{2}\right)^{2t}$$

$$9500 = 5000 (1.03375)^{2t}$$

$$\frac{9500}{5000} = \frac{5000 (1.068639063t)}{5000}$$

$$\frac{1.9}{1.068639063} = \frac{1.068639063t}{1.068639063}$$

$$1.777962332 = t$$

$$\sqrt{1.8 \text{ years}}$$

½ out of 3

+ ½ mark for substitution

Exemplar 4

$$\begin{aligned} A &= \$9500 & A &= P \left(1 + \frac{r}{n}\right)^{nt} \\ P &= \$5000 & 9500 &= 5000 \left(1 + \frac{0.675}{2}\right)^{2t} \\ n &= 2 & 1.9 &= (1.3375)^{2t} \\ t &= ? & \log 1.9 &= 2t \log 1.3375 \\ r &= 0.675 & \log 1.9 &= t(2 \log 1.3375) \\ & & \frac{\log 1.9}{2 \log 1.3375} &= t \\ & & 1.104 &= t \end{aligned}$$

It will take Loïc 1.104 years to grow to \$9500.

2½ out of 3

- + 1 mark for applying logarithms
- + 1 mark for power law
- + ½ mark for evaluating a quotient of logarithms

Question 5

T5

Solve, algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta - \csc \theta - 6 = 0$$

Solution

Method 1

$$(\csc \theta - 3)(\csc \theta + 2) = 0$$

$$\csc \theta = 3$$

$$\csc \theta = -2$$

1 mark for solving for $\csc \theta$

$$\sin \theta = \frac{1}{3}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for reciprocal

$$\theta_r = 0.339\ 836\dots$$

$$\theta = 0.340; 2.802$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

Method 2

$$\frac{1}{\sin^2 \theta} - \frac{1}{\sin \theta} - 6 = 0$$

1 mark for reciprocal

$$1 - \sin \theta - 6 \sin^2 \theta = 0$$

$$6 \sin^2 \theta + \sin \theta - 1 = 0$$

$$(3 \sin \theta - 1)(2 \sin \theta + 1) = 0$$

$$\sin \theta = \frac{1}{3}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for solving for $\sin \theta$

$$\theta_r = 0.339\ 836\dots$$

$$\theta = 0.340; 2.802$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

Exemplar 1

$$(\csc \theta + 3)(\csc \theta - 2) = 0$$

$$\begin{array}{ll} \csc \theta = -3 & \csc \theta = 2 \\ \sin \theta = -\frac{1}{3} & \sin \theta = \frac{1}{2} \\ & \theta = \pi/6 \end{array}$$

2 out of 4

+ 1 mark for solving for $\csc \theta$

+ 1 mark for reciprocal

+ ½ mark for consistent value of θ

– ½ mark for procedural error (line 1)

Exemplar 2

$$\csc^2 \theta - \csc \theta - 6 = 0$$

$$x^2 - x - 6$$

$$(x+2)(x-3)$$

$$\csc \theta = -2, 3$$

$$\sin = -\frac{1}{2} \rightarrow Q_3, Q_4$$

$$Q_3 \rightarrow \frac{7\pi}{6}$$

$$Q_4 \rightarrow \frac{11\pi}{6}$$

$$\sin = \frac{1}{3} \rightarrow Q_1, Q_2$$

$$\sin^{-1}\left(\frac{1}{3}\right) = 0.3398\dots$$

$$Q_1 \rightarrow 0.340$$

$$Q_2 \rightarrow \pi - 0.3398\dots = 2.802$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, 0.340, 2.802$$

4 out of 4

award full marks

E3 (variable introduced without being defined in line 2)

E2 (changing an equation to an expression in lines 2 and 3)

E3 (variable omitted in an equation in line 5)

Exemplar 3

$$x = \csc \theta$$

$$x^2 - x - 6 = 0$$

$$(x - 3)(x + 2) = 0$$

$$x = 3 \quad x = -2$$

$$\csc \theta = 3 \quad \csc \theta = -2$$

$$\sin \theta = \frac{1}{3} \quad \sin \theta = -\frac{1}{2}$$

$$\theta_r = 19.471.. \quad \theta_r = 30$$

$$\theta_1 = 19.471 \quad \theta_3 = 30$$

$$\theta_2 = 199.471 \quad \theta_4 = 210$$

3 out of 4

+ 1 mark for solving for $\csc \theta$

+ 1 mark for reciprocal

+ 1 mark for two correct values of θ

E5 (units of measure omitted in final answer)

E5 (answer stated in degrees instead of radians)

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Question 6

P1

Determine how many 4-digit numbers can be made using the digits 0, 1, 2, 3, 4, and 5, if the third digit must be 3, and repetition is not allowed.

Solution

$$\frac{4}{\text{not 0 or 3}} \cdot \frac{4}{\quad} \cdot \frac{1}{3} \cdot \frac{3}{\quad} = 48$$

½ mark for restriction of first digit

½ mark for restriction of third digit

1 mark for fundamental counting principle

2 marks

Exemplar 1

$$5 \cdot 4 \cdot 1 \cdot 3 = 60$$

$${}_5P_3 = 60$$

60 4-digit numbers can be made.

1½ out of 2

+ ½ mark for restriction of third digit

+ 1 mark for fundamental counting principle

Exemplar 2

0, 1, 2, 3 4, 5

5! only 3 spots

$$5 \cdot 4 \cdot 3$$

60 numbers can be made using
this outline.

1½ out of 2

+ ½ mark for restriction of third digit

+ 1 mark for fundamental counting principle

Exemplar 3

$$\begin{array}{ccccccc} \underline{4} & \times & \underline{4} & \times & \underline{1} & \times & \underline{2} & = & \boxed{32 \text{ numbers}} \\ \uparrow & & & & & & & & \\ \text{Zero can't} & & & & & & & & \\ \text{go Here} & & & & & & & & \end{array}$$

1½ out of 2

award full marks

– ½ mark for procedural error (incorrect last value)

Question 7

R3, R5

Describe the transformations required to obtain the graph of the function $y = -4f\left(\frac{1}{3}x\right)$ from the graph of $y = f(x)$.

Solution

- | | |
|---------------------------------------|--------------------------------|
| • Reflection over the x -axis | 1 mark for vertical reflection |
| • Vertical stretch by a factor of 4 | 1 mark for vertical stretch |
| • Horizontal stretch by a factor of 3 | 1 mark for horizontal stretch |

3 marks

Exemplar 1

Vertical reflection
horizontal stretch by a factor of $\frac{1}{3}$
Vertical stretch by a factor of 4.

2 out of 3

+ 1 mark for vertical reflection
+ 1 mark for vertical stretch

Exemplar 2

- Flip over the x-axis
 - A vertical stretch
 - A horizontal stretch
-

1 out of 3

+ 1 mark for vertical reflection

Exemplar 3

A vertical stretch by a factor of 4 followed by a reflection in the x-axis and a horizontal compression of factor $\frac{1}{3}$.

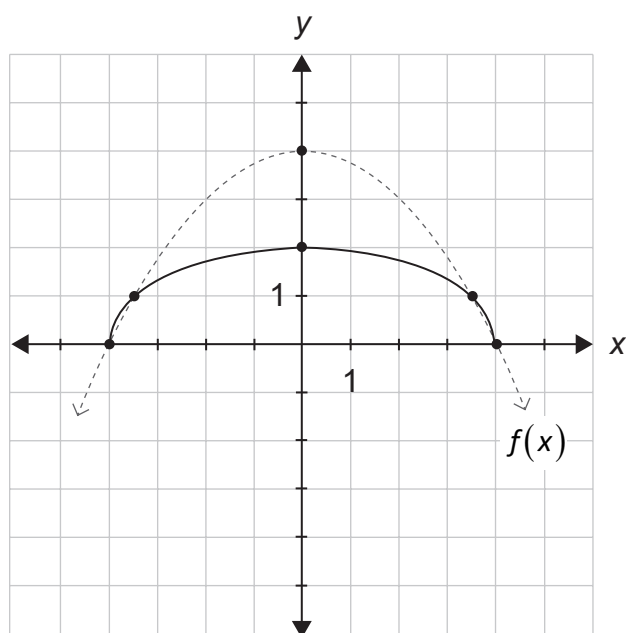
3 out of 3

Question 8

R13

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.

Solution



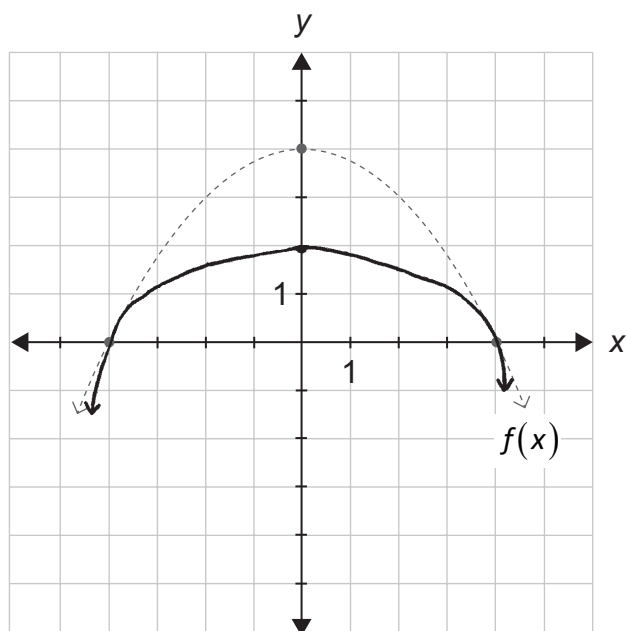
1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant points,
 $\{0 \leq y \leq 1\}$

$\frac{1}{2}$ mark for shape above invariant points,
 $\{1 \leq y \leq 2\}$

2 marks

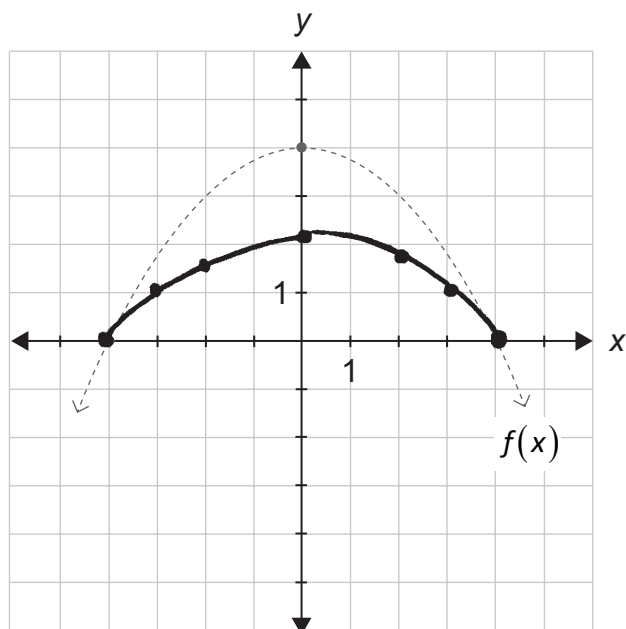
Exemplar 1



½ out of 2

+ ½ mark for shape above invariant points $\{1 \leq y \leq 2\}$

Exemplar 2



1½ out of 2

+ 1 mark for restricting domain

+ ½ mark for shape above invariant points $\{1 \leq y \leq 2\}$

Question 9

R11

Describe how to determine if $(x - a)$ is a factor of the polynomial function, $p(x)$.

Solution

$(x - a)$ is a factor of $p(x)$, if and only if $p(a) = 0$.

1 mark

Exemplar 1

$x-a$ would be a factor
if the remainder would
be 0.

1 out of 1

Exemplar 2

Plug $x-a$ into and if there
is no remainder it is a factor.
 $\downarrow$
 $\overline{a) \quad 3 \quad 1 \quad 6 \quad 10}$

$\frac{1}{2}$ out of 1

award full marks

– $\frac{1}{2}$ mark for lack of clarity in description

Exemplar 3

If $p(a)=0$, then $(x-a)$
is a factor.

1 out of 1

Question 10

R8

Express $\frac{1}{2}\log x - 2\log y + \log z$ as a single logarithm.

Solution

$$\log\left(\frac{\sqrt{x} \cdot z}{y^2}\right)$$

1 mark for product law

1 mark for quotient law

1 mark for power law ($\frac{1}{2}$ mark for each)

3 marks

Exemplar 1

$$\log\left(\frac{\frac{1}{2}x}{2y}\right)(z) \quad \boxed{\log\left(\frac{\frac{1}{2}xz}{2y}\right)}$$

2 out of 3

- + 1 mark for product law
- + 1 mark for quotient law

Exemplar 2

$$\log\left(\frac{x^{-2}}{y^2 \cdot z}\right)$$

1½ out of 3

- + 1 mark for quotient law
- + ½ mark for power law

Exemplar 3

$$\log\left(\frac{\sqrt{x}+z}{y^2}\right)$$

1½ out of 3

- + 1 mark for quotient law
- + ½ mark for power law

Exemplar 4

$$\begin{aligned} & \frac{1}{2}\log x - 2\log y + \log z \\ & \log x^{1/2} - \log y^2 + \log z \\ & = \left(\frac{\log x^{1/2}}{\log y^2}\right)(\log z) \end{aligned}$$

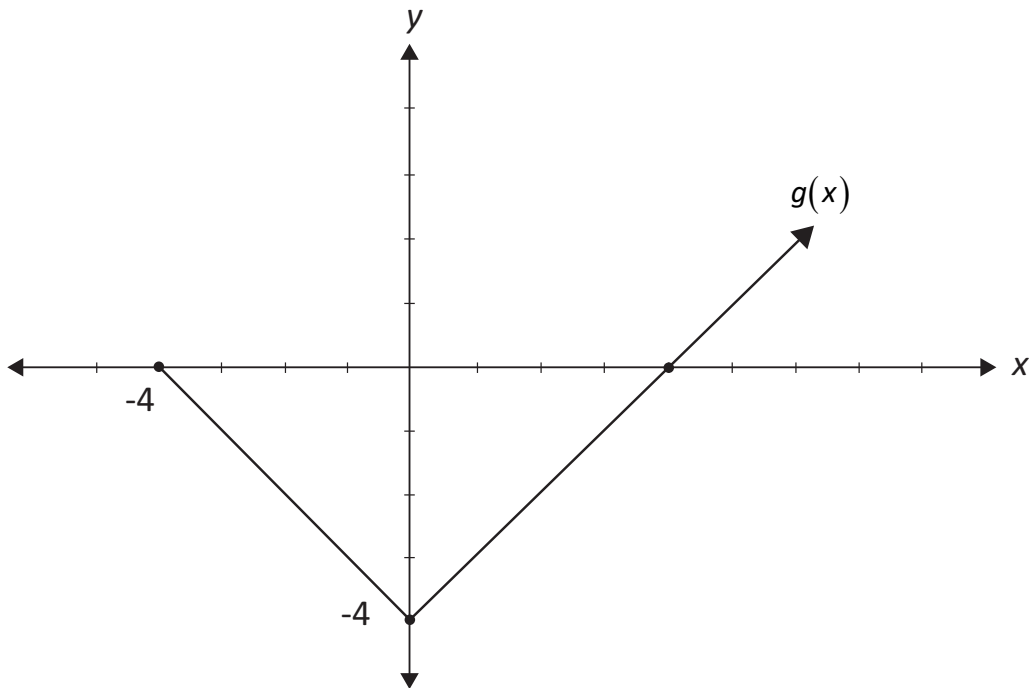
2 out of 3

- award full marks
- 1 mark for concept error (not writing as a single logarithm)

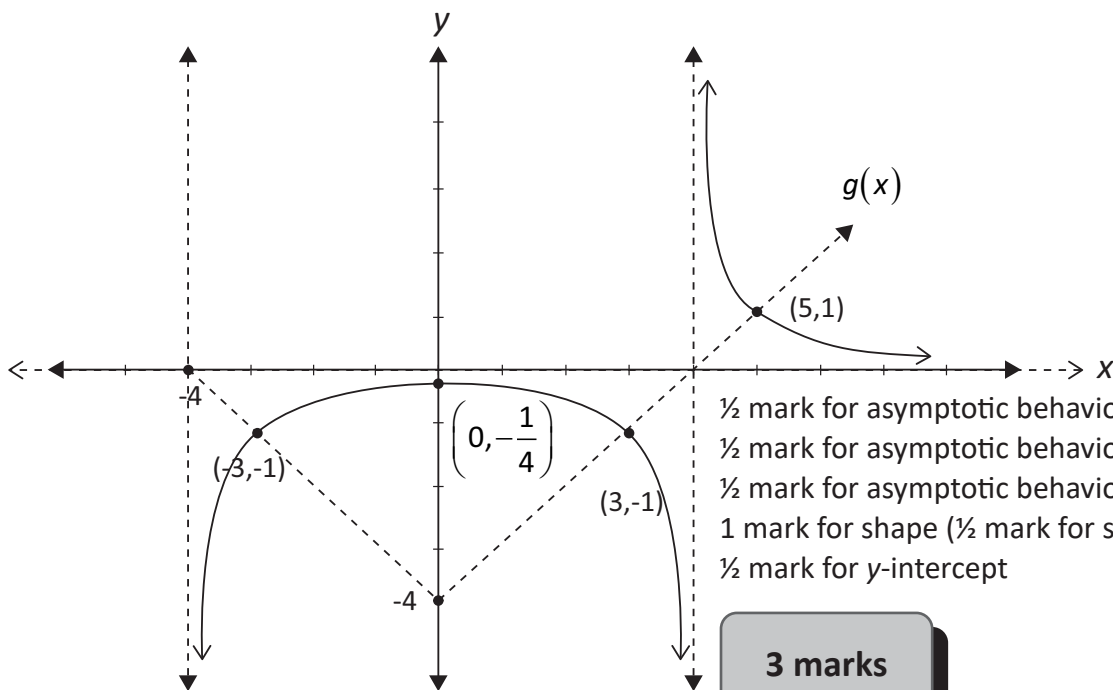
Question 11

R1

Given the graph of $g(x)$, sketch the graph of $f(x) = \frac{1}{g(x)}$.



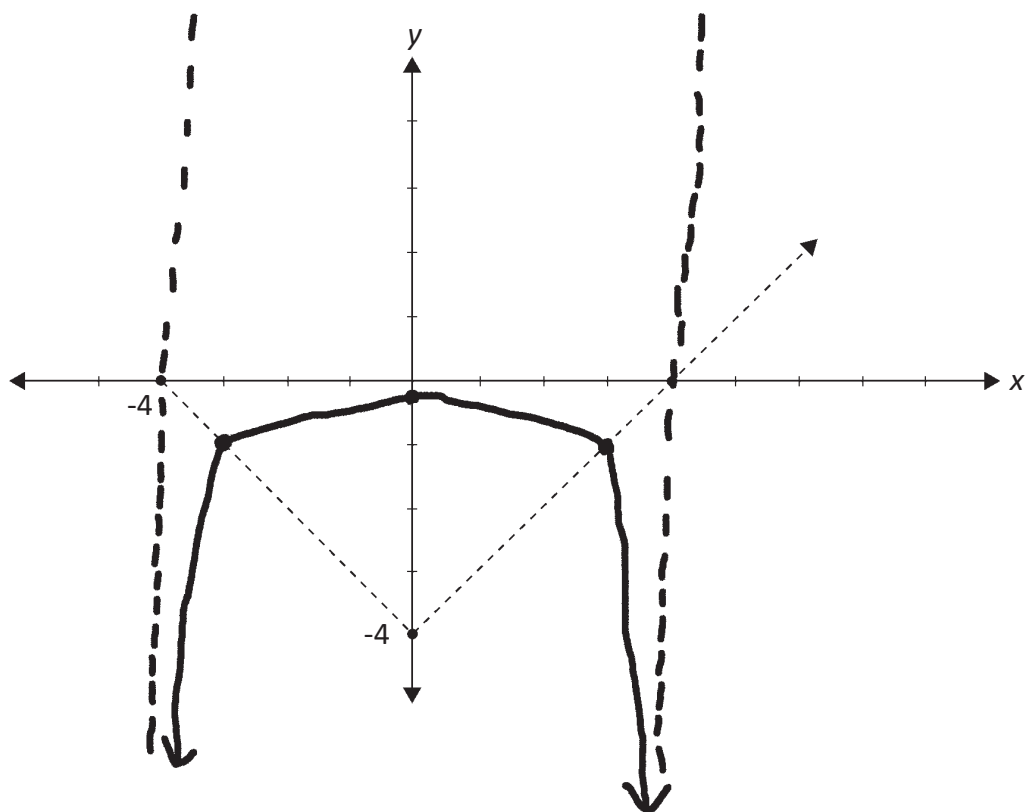
Solution



$\frac{1}{2}$ mark for asymptotic behaviour approaching $x = -4$
 $\frac{1}{2}$ mark for asymptotic behaviour approaching $x = 4$
 $\frac{1}{2}$ mark for asymptotic behaviour approaching $y = 0$
 1 mark for shape ($\frac{1}{2}$ mark for shape in each section)
 $\frac{1}{2}$ mark for y-intercept

3 marks

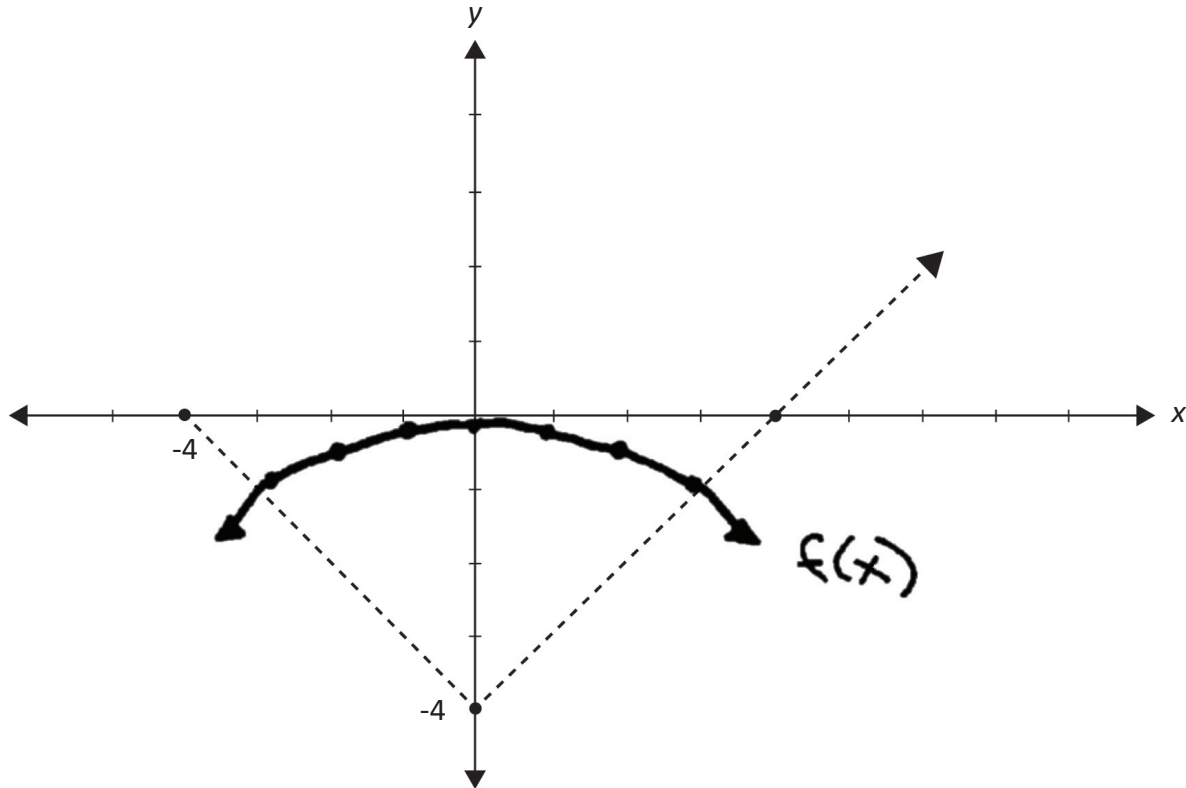
Exemplar 1



2 out of 3

- + ½ mark for asymptotic behaviour approaching $x = -4$
- + ½ mark for asymptotic behaviour approaching $x = 4$
- + ½ mark for shape
- + ½ mark for y -intercept

Exemplar 2

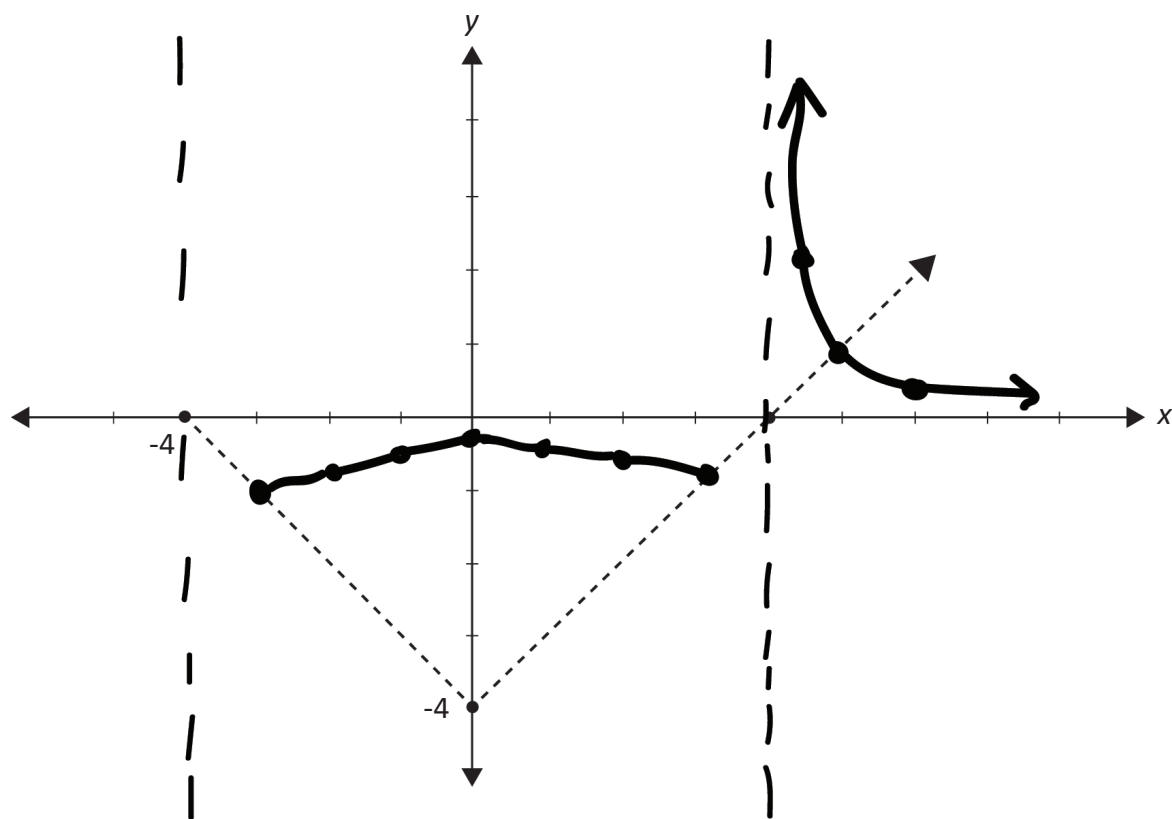


1 out of 3

+ ½ mark for shape

+ ½ mark for y-intercept

Exemplar 3



2 out of 3

- + ½ mark for asymptotic behaviour approaching $x = 4$
- + ½ mark for asymptotic behaviour approaching $y = 0$
- + ½ mark for shape
- + ½ mark for y -intercept
- E10 (horizontal asymptote omitted but still implied)

Question 12

T6

Given the identity $\frac{1 - \tan^2 \theta}{\sec^2 \theta} = \cos 2\theta$,

- state a non-permissible value of θ .
- prove the identity for all permissible values of θ .

Solution

a) Any value where $\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$ or $\theta = 90^\circ + 180^\circ k, k \in \mathbb{Z}$, is acceptable.

1 mark

b)

Method 1

Left-Hand Side	Right-Hand Side
$ \begin{array}{r} 1 - \frac{\sin^2 \theta}{\cos^2 \theta} \\ \hline \frac{1}{\cos^2 \theta} \\ \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \\ \hline \frac{1}{\cos^2 \theta} \\ \frac{\cos 2\theta}{\cancel{\cos^2 \theta}} \\ \hline \frac{1}{\cancel{\cos^2 \theta}} \\ \cos 2\theta \end{array} $	$\cos 2\theta$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\frac{1}{\sec^2 \theta} - \frac{\tan^2 \theta}{\sec^2 \theta}$ $\cos^2 \theta - \frac{\sin^2 \theta}{\cancel{\cos^2 \theta} \cdot \frac{1}{\cancel{\cos^2 \theta}}}$ $\cos^2 \theta - \sin^2 \theta$ $\cos 2\theta$	$\cos 2\theta$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Exemplar 1

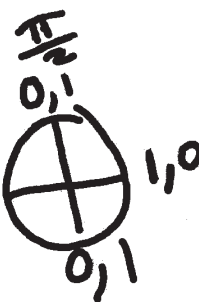
a)

$$\boxed{\theta \neq \frac{\pi}{2}}$$

$$(\cos \theta)(\cos \theta)$$

$$\cos \theta \neq 0$$

$$\theta \neq \frac{\pi}{2}$$



1 out of 1

b)

Left-Hand Side	Right-Hand Side
	$\frac{1}{\sec^2 \theta} = \frac{\cos^2 \theta}{1}$
	$\cos 2\theta = \frac{1 - \tan^2 \theta}{\sec^2 \theta}$
	$= (1 - \tan^2 \theta)(\cos^2 \theta)$
	$= \cos^2 \theta - \tan^2 \theta \cos^2 \theta$
	$= \cos^2 \theta - \frac{\sin^2 \theta \cancel{\cos^2 \theta}}{\cancel{\cos^2 \theta}}$
	$= \cos^2 \theta - \sin^2 \theta$
	$\boxed{\cos 2\theta = \cos 2\theta}$

3 out of 3

award full marks

E2 (equating the two sides when proving an identity)

Exemplar 2

a)

$$\theta \neq \pi/2, 3\pi/2, \pi, 2\pi$$

0 out of 1

b)

Left-Hand Side	Right-Hand Side
$\begin{aligned} &= \frac{1 - \frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{1}{\cos^2 \theta}} \\ &= \frac{\cos^2 \theta - \frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{1}{\cos^2 \theta}} \\ &= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \cdot \frac{\cos^2 \theta}{1} \\ &= \frac{2\cos^2 \theta - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta} \\ &= \cos^2 \theta - \sin^2 \theta \cos^2 \theta \\ &= \cos^2 \theta (1 - \sin^2 \theta) \\ &= \cos^2 \theta \cos^2 \theta \\ &= 2\cos^2 \theta - 1 \checkmark \end{aligned}$	$= 2\cos^2 \theta - 1 \checkmark$

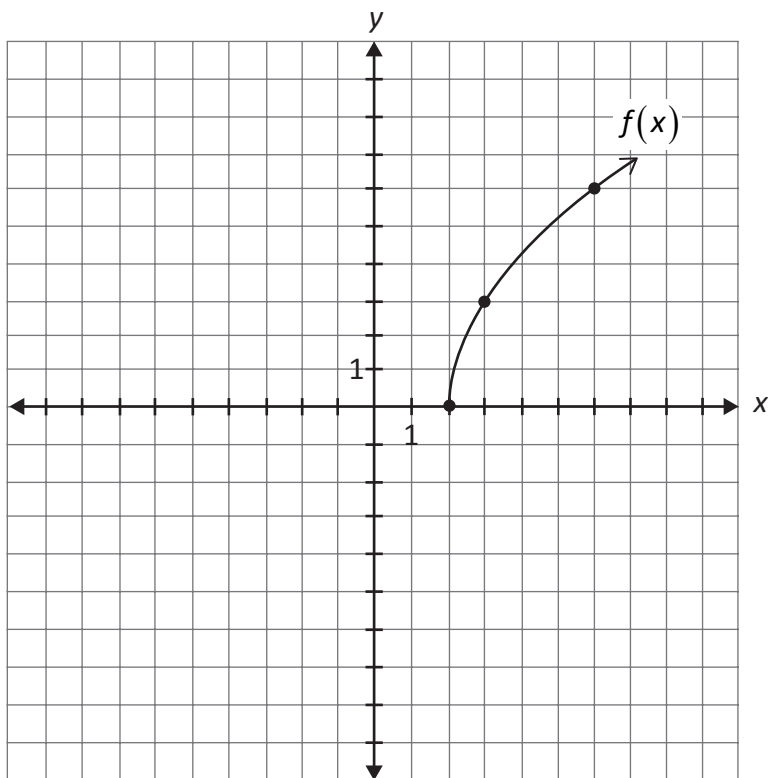
1 out of 3

+ 1 mark for correct substitution of identities

Question 13

R13

Determine an equation of the radical function represented by the graph of $f(x)$.



Solution

$$f(x) = \underline{3\sqrt{x-2}}$$

1 mark for vertical stretch

1 mark for horizontal translation

2 marks

Exemplar 1

$$f(x) = 3\sqrt{x+2}$$

1 out of 2

+ 1 mark for vertical stretch

Exemplar 2

$$f(x) = 3f\sqrt{x-2}$$

1½ out of 2

award full marks

– ½ mark for procedural error (including f)

Exemplar 3

$$f(x) = 3f(x-2)$$

1 out of 2

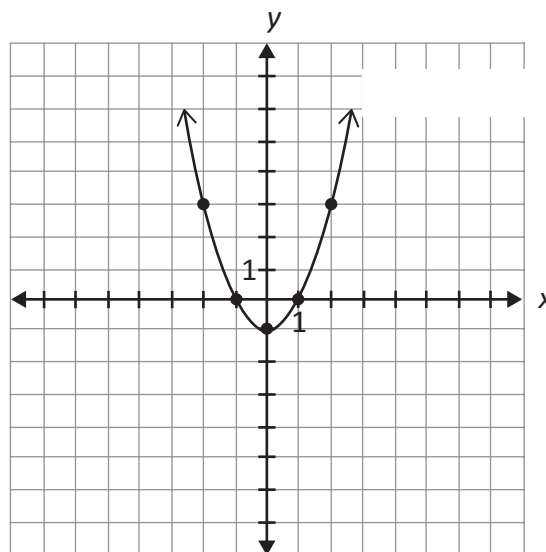
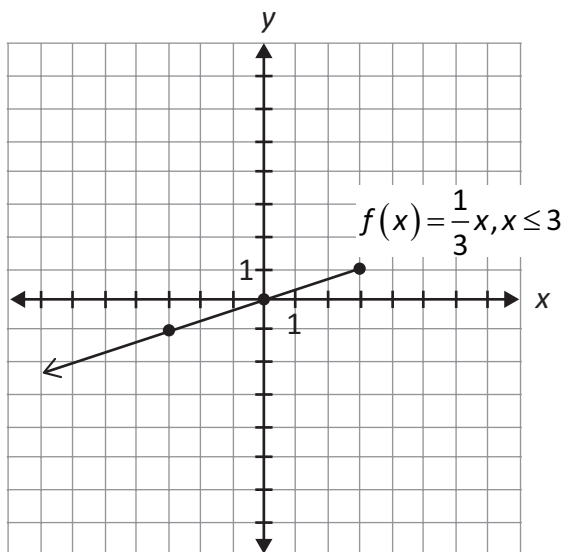
award full marks

– 1 mark for concept error (including f instead of radical)

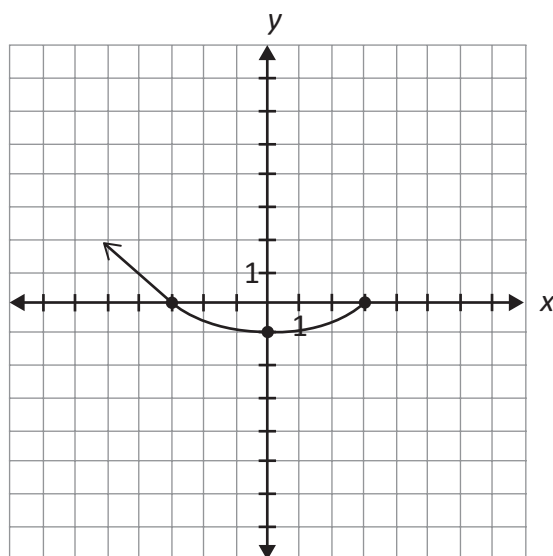
Question 14

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(f(x))$.



Solution

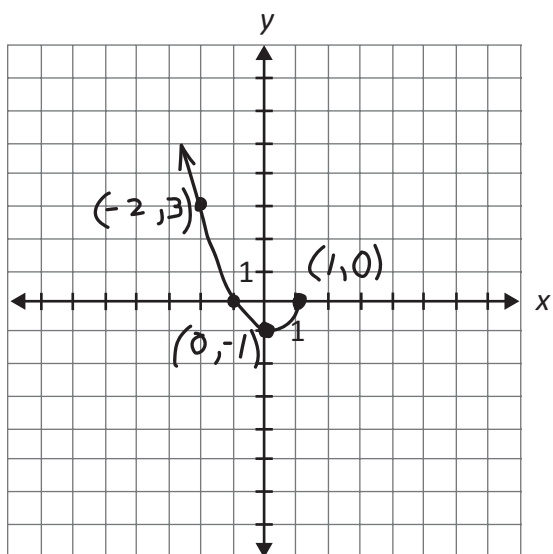


1 mark for $g(f(x))$

1 mark for restricted domain

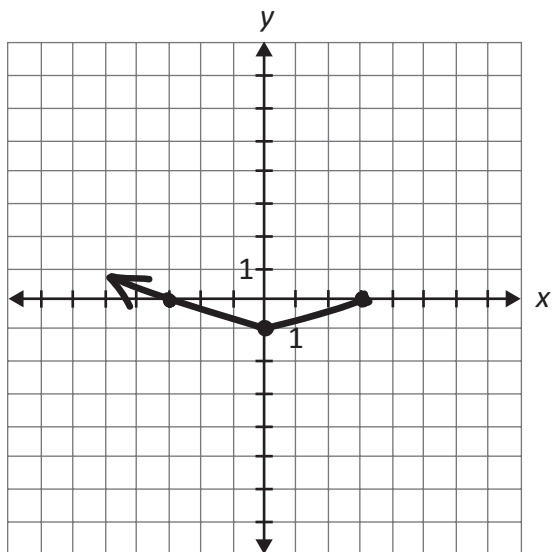
2 marks

Exemplar 1



0 out of 2

Exemplar 2



1½ out of 2

award full marks

– ½ mark for incorrect shape of graph

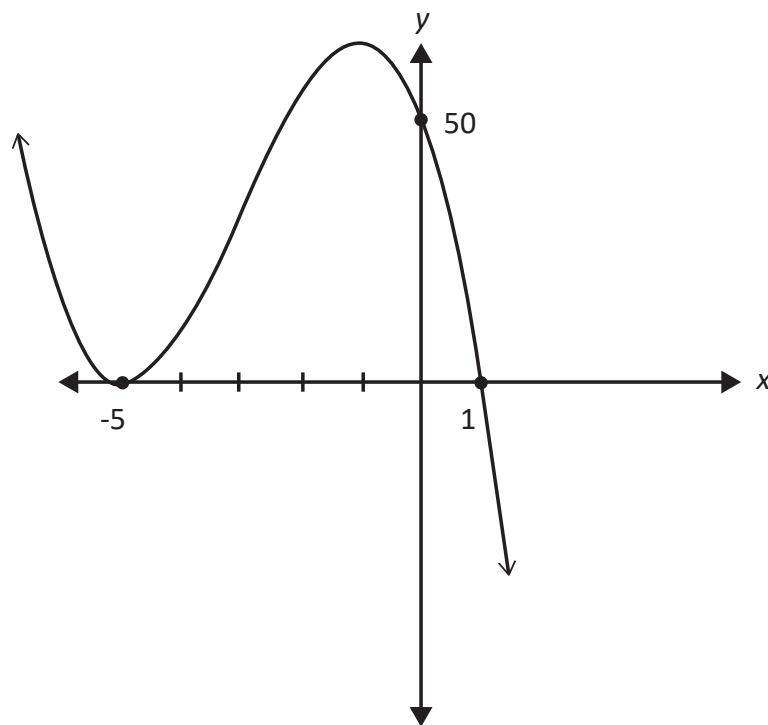
Question 15

R12

Sketch a graph of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 3
- $p(x)$ has a zero at -5 with a multiplicity of 2
- $p(x)$ has a zero at 1
- $p(x)$ has a leading coefficient of -2

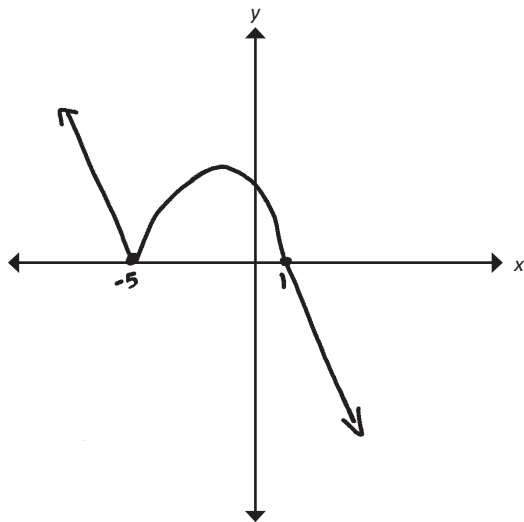
Solution



$\frac{1}{2}$ mark for x-intercepts
1 mark for multiplicity of 2 at $x = -5$
 $\frac{1}{2}$ mark for y-intercept
 $\frac{1}{2}$ mark for end behaviour
 $\frac{1}{2}$ mark for shape of cubic function

3 marks

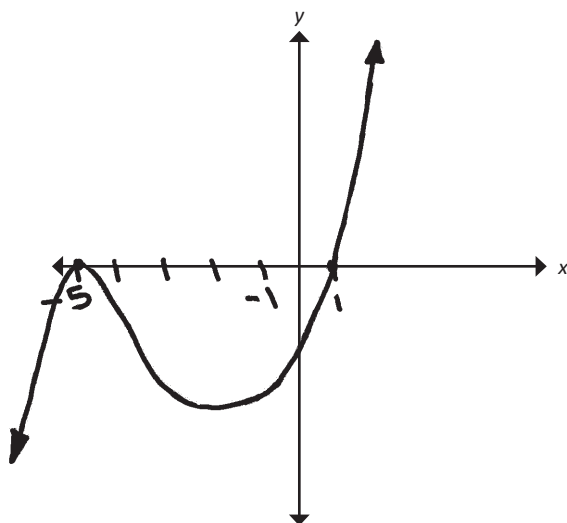
Exemplar 1



2 out of 3

- + ½ mark for x-intercepts
- + 1 mark for multiplicity of 2 at $x = -5$
- + ½ mark for end behaviour

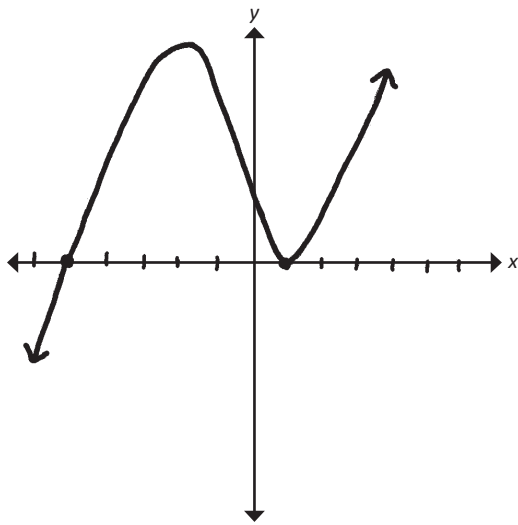
Exemplar 2



2 out of 3

- + ½ mark for x-intercepts
- + 1 mark for multiplicity of 2 at $x = -5$
- + ½ mark for shape of cubic function

Exemplar 3



1 out of 3

+ ½ mark for x-intercepts

+ ½ mark for shape of cubic function

E9 (scale values on axes not indicated)

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Question 16

R14

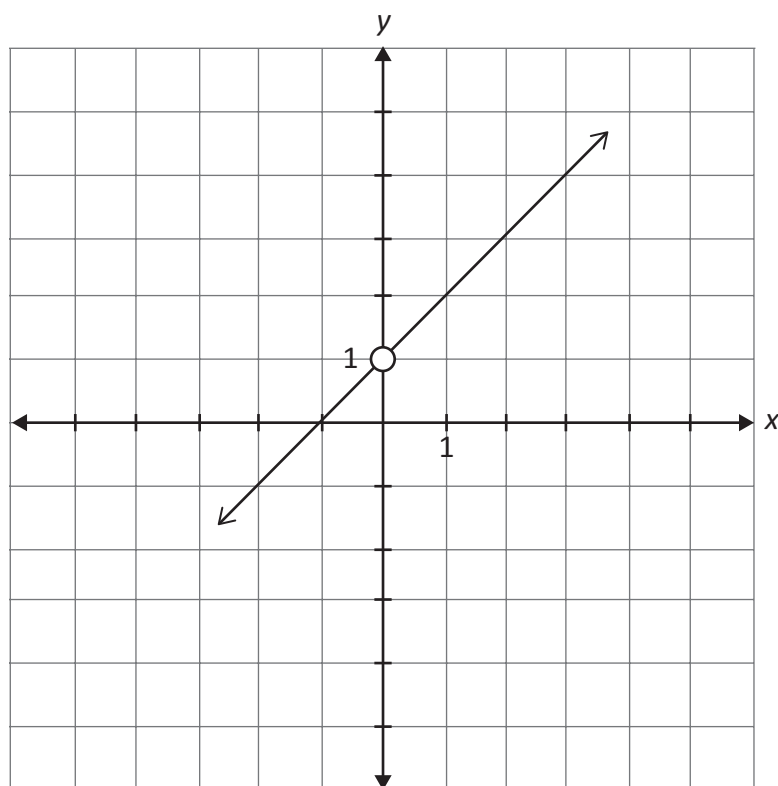
Sketch the graph of $f(x) = \frac{x^2 + x}{x}$.

Solution

$$f(x) = \frac{\cancel{x}(x+1)}{\cancel{x}}$$

$$f(x) = x + 1, x \neq 0$$

$\therefore$ there is a point of discontinuity (hole) at $(0,1)$.

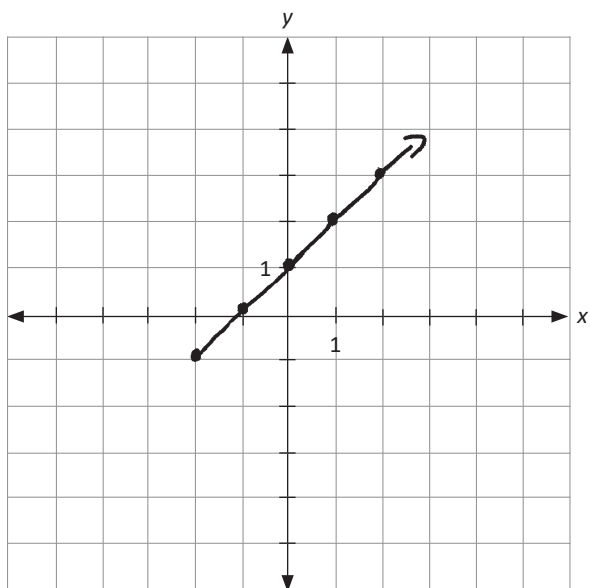


1 mark for point of discontinuity (hole) at $x = 0$

1 mark for shape of $y = x + 1$

2 marks

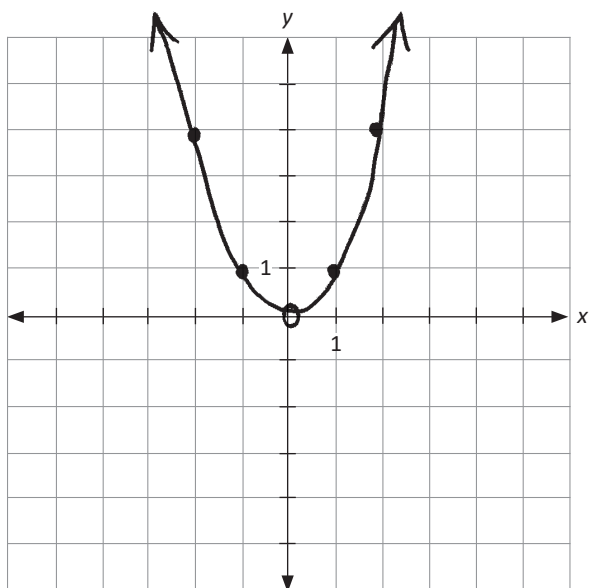
Exemplar 1



1 out of 2

+ 1 mark for shape of $y = x + 1$
E9 (arrowhead omitted)

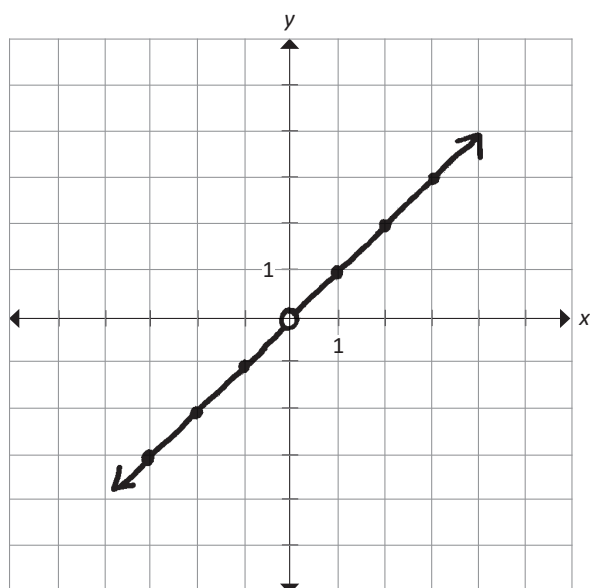
Exemplar 2



1 out of 2

+ 1 mark for point of discontinuity (hole) at $x = 0$

Exemplar 3

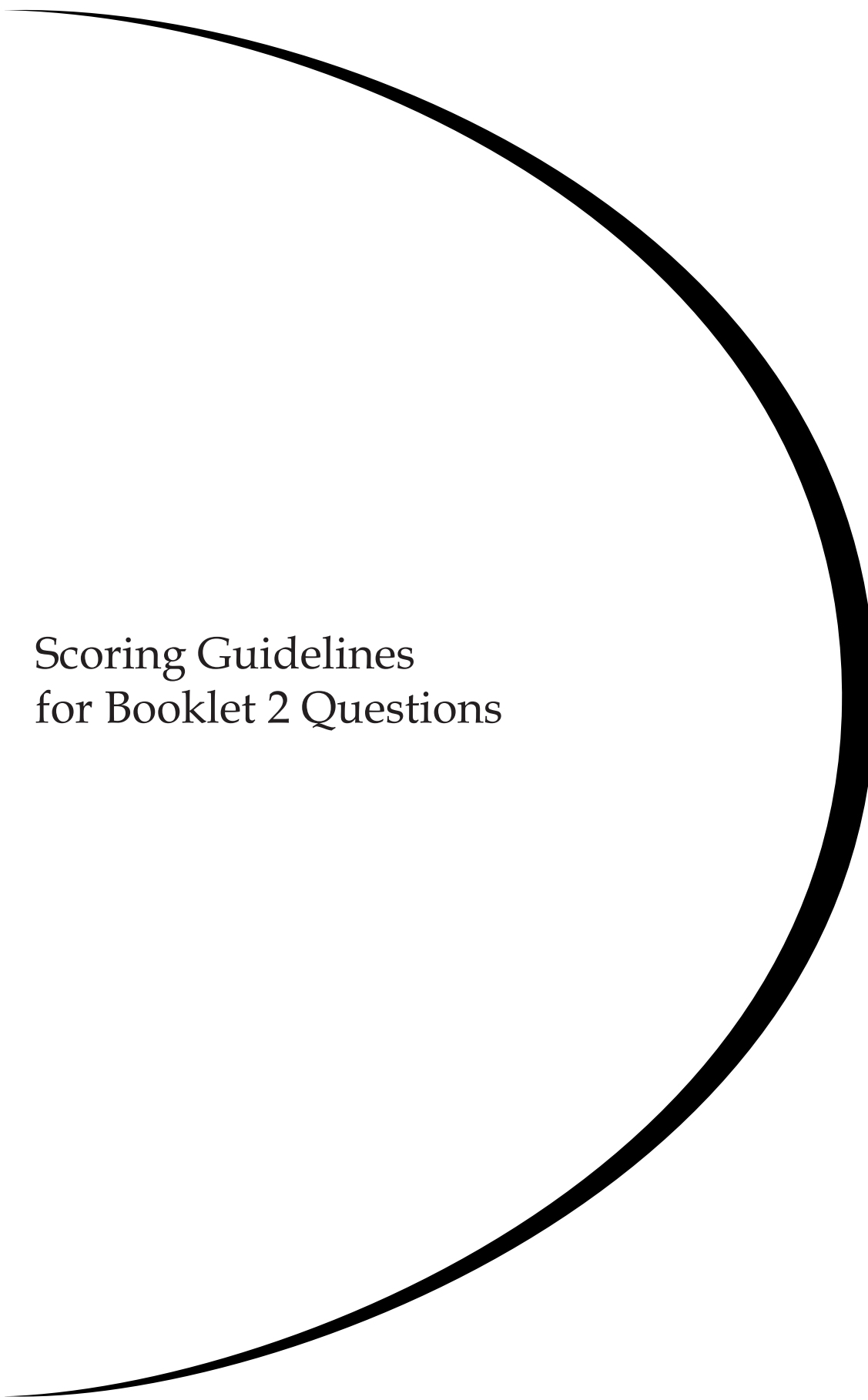


$$f(x) = \frac{x^2 + x}{x}$$

$$f(x) = x, x \neq 0$$

1 out of 2

+ 1 mark for point of discontinuity (hole) at $x = 0$



Scoring Guidelines for Booklet 2 Questions

Answer Key for Selected Reponse Questions

Question	Answer	Learning Outcome
17	B	R2
18	A	T3
19	D	R7
20	A	T3
21	B	P4
22	D	T5
23	C	R12
24	D	T6
25	B	R1

Question 17

R2

Identify the function that has a different range than $f(x) = x^2$.

a. $y = 4f(x)$

b. $y = f(x) + 4$

c. $y = f(x + 4)$

d. $y = f(-x)$

Question 18

T3

Identify the exact value of $\sin\left(-\frac{7\pi}{3}\right)$.

a. $-\frac{\sqrt{3}}{2}$

b. $-\frac{1}{2}$

c. $\frac{1}{2}$

d. $\frac{\sqrt{3}}{2}$

Question 19

R7

Identify the value of a in the equation, $\log_a 3 = \frac{1}{4}$.

a. $\frac{1}{64}$

b. $3^{\frac{1}{4}}$

c. 12

d. 81

Question 20**T3**

Identify a value of x where the graphs of $y = \cos x$ and $y = \sin x$ intersect.

a. $-\frac{3\pi}{4}$

b. $-\frac{\pi}{4}$

c. 0

d. $\frac{3\pi}{4}$

Question 21**P4**

Identify the number of negative terms in the binomial expansion of $(2x - y)^6$.

a. 0

b. 3

c. 4

d. 6

Question 22**T5**

Identify the general solution of the equation, $\sin x = 1$.

a. $x = k\pi, k \in \mathbb{Z}$

b. $x = 2k\pi, k \in \mathbb{Z}$

c. $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

d. $x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$

Question 23**R12**

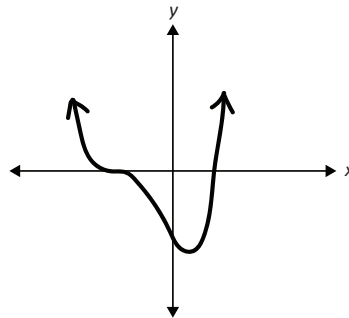
Identify a possible degree of the polynomial function represented below.

a. 2

b. 3

c. 4

d. 5

**Question 24****T6**

Identify the expression that is equivalent to $\cos x \cot x$.

a. $\frac{1 - \sin x}{\sin x}$

b. $\frac{\sin^2 x - 1}{\sin x}$

c. $\sin x$

d. $\frac{1 - \sin^2 x}{\sin x}$

Question 25**R1**

Given that the point $(-7, -5)$ is on the graph of $y = g(x)$, identify the corresponding point on the graph of $y = |g(x)|$.

a. $(7, 5)$

b. $(-7, 5)$

c. $(7, -5)$

d. $(-5, -7)$

Question 26

R11

Given that $x = 1$ is one of the zeros of $p(x) = x^3 + x^2 - 10x + 8$, express $p(x)$ in completely factored form.

Solution

$$p(x) = x^3 + x^2 - 10x + 8$$

1		1	1	-10	8
		↓	1	2	-8
		1	2	-8	0

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x - 1)(x^2 + 2x - 8)$$

$$p(x) = (x - 1)(x + 4)(x - 2)$$

1 mark for consistent factors

2 marks

Exemplar 1

$$\begin{array}{r|rrrr} 1 & 1 & 1 & -10 & 8 \\ & \downarrow & 1 & 2 & 8 \\ \hline & 1 & 2 & -8 & 0 \end{array}$$

$$\begin{aligned} p(x) &= (x-1)(x^2+2x-8) \\ &= (x-1)(x+4)(x-2) \end{aligned}$$

zeros

$$x = -4$$
$$x = 1$$
$$x = 2$$

1½ out of 2

award full marks

– ½ mark for procedural error (solving for zeros)

Exemplar 2

$$\begin{array}{r} x^2 + 2x - 8 \\ x-1 \overline{) x^3 + x^2 - 10x + 8} \\ \underline{+ x^3 + x^2} \\ 2x^2 - 10x \\ \underline{+ 2x^2 + 2x} \\ -8x + 8 \\ \underline{+ 8x - 8} \\ 0 \end{array}$$

$$\begin{aligned} &x^2 + 2x - 8 \\ &(x+4)(x-2) \end{aligned}$$

$$p(x) = (x-1)(x+4)(x-2)$$

2 out of 2

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Question 27

T3

Determine the exact value of $\frac{\csc^2 \theta - 1}{1 - \cos^2 \theta}$ if $\theta = \frac{5\pi}{6}$.

Solution

Method 1

$$\frac{\csc^2\left(\frac{5\pi}{6}\right) - 1}{1 - \cos^2\left(\frac{5\pi}{6}\right)}$$

$$\frac{(2)^2 - 1}{1 - \left(-\frac{\sqrt{3}}{2}\right)^2}$$

$$\frac{4 - 1}{1 - \frac{3}{4}}$$

$$\frac{3}{\frac{1}{4}}$$

$$12$$

1 mark for $\csc\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

1 mark for $\cos\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

2 marks

Method 2

$$\frac{\cot^2 \theta}{\sin^2 \theta}$$

$$\frac{\cot^2\left(\frac{5\pi}{6}\right)}{\sin^2\left(\frac{5\pi}{6}\right)}$$

$$\frac{(-\sqrt{3})^2}{\left(\frac{1}{2}\right)^2}$$

$$\frac{3}{\frac{1}{4}}$$

$$12$$

1 mark for $\cot\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

1 mark for $\sin\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

2 marks

Exemplar 1

$$\frac{\csc^2 \theta - 1}{1 - \cos^2 \theta}$$

$$\frac{\cot^2 \theta}{\sin^2 \theta}$$

$$\frac{\frac{\cos^2 \theta}{\sin^2 \theta}}{\sin^2 \theta}$$

$$\frac{\cos^2 \theta}{\sin^2 \theta} \cdot \cancel{\sin^2 \theta}$$

$$\cos^2 \theta$$

$$\left(\cos \frac{5\pi}{6}\right)^2$$

$$\left(\frac{-\sqrt{3}}{2}\right)^2$$

$$\frac{3}{4}$$

1 out of 2

+ 1 mark for $\cos\left(\frac{5\pi}{6}\right)$

Exemplar 2

$$\frac{\cot^2 \theta}{\sin^2 \theta} = \frac{\cos^2 \theta}{\sin^2 \theta} \div \sin^2 \theta$$

$$= \frac{\frac{-\sqrt{3}}{2}}{\frac{1}{2}} \times \frac{2}{1}$$

$$= \frac{-\sqrt{3}}{\cancel{2}} \cdot \frac{\cancel{2}}{1} \times \left(\frac{2}{1}\right)$$

$$= \frac{-\sqrt{3}}{1} \cdot \frac{2}{1}$$

$$= -2\sqrt{3}$$

1½ out of 2

award full marks

– ½ mark for procedural error (not squaring the values)

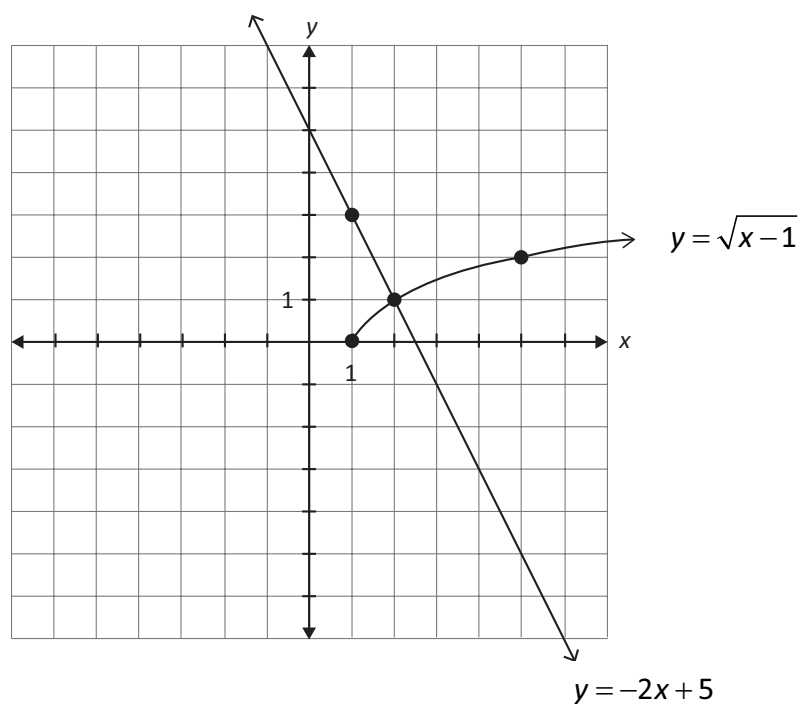
Question 28

R13

Solve, graphically.

$$-2x + 5 = \sqrt{x - 1}$$

Solution



$$\therefore x = 2$$

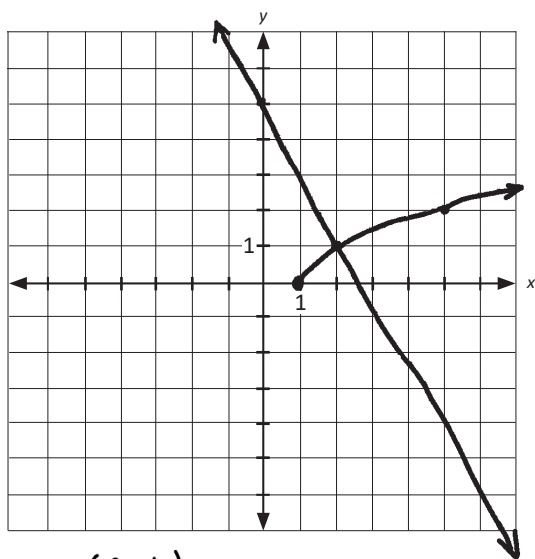
1 mark for shape of a radical function

1 mark for horizontal translation

1 mark for x-value consistent with intersection point of the two graphs

3 marks

Exemplar 1



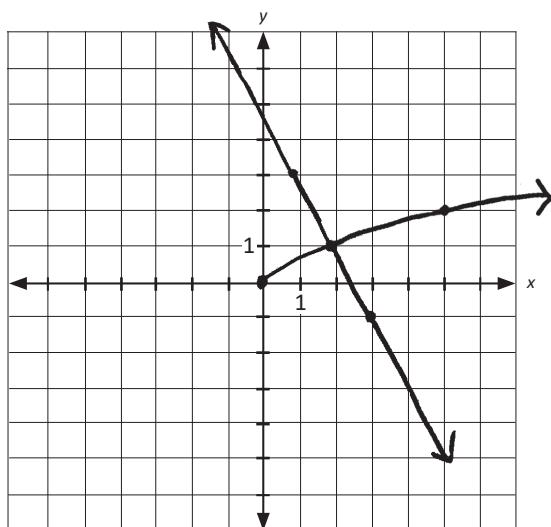
$(2, 1)$

2 out of 3

+ 1 mark for shape of a radical function

+ 1 mark for horizontal translation

Exemplar 2



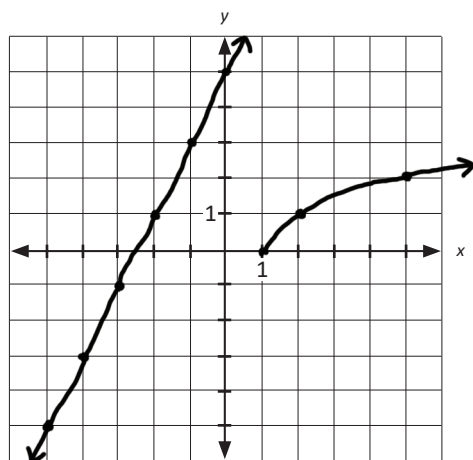
$x = 2$

2½ out of 3

award full marks

– ½ mark for procedural error (one incorrect point on radical graph)

Exemplar 3



This equation has
no solutions
because the lines
don't intersect

2½ out of 3

award full marks

– ½ mark procedural error (incorrect linear graph)

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Question 29

R8

If $\log_a 2 = p$ and $\log_a 5 = q$, express $\log_a 40$ in terms of p and q .

Solution

$$\begin{aligned}\log_a 40 &= \log_a (2^3 \cdot 5) \\ &= 3\log_a 2 + \log_a 5 \\ &= 3p + q\end{aligned}$$

1 mark for power law

1 mark for product law

2 marks

Exemplar 1

$$\log_a 40 = \log_a 5 + \log_a 2 + \log_a 2 + \log_a 2$$

$$\log_a 40 = q + p + p + p$$

$$\log_a 40 = qp^3$$

1½ out of 2

award full marks

– ½ mark for arithmetic error in line 3

Exemplar 2

$$2 \cdot 2 \cdot 2 \cdot 5 = 40$$

$$p \cdot p \cdot p \cdot q$$

$$\log_a 40 = p \cdot p \cdot p \cdot q$$

$$\log_a 40 = p^3 q$$

0 out of 2

Question 30

R6

Describe how to sketch the graph of $y=f^{-1}(x)$, given the graph of a linear function, $y=f(x)$.

Solution

The graph of $y=f^{-1}(x)$ is obtained by reflecting the graph of $y=f(x)$ over the line $y=x$.

or

To sketch the inverse, $y=f^{-1}(x)$, transform all points on the graph of $y=f(x)$ from (x,y) to (y,x) .

1 mark

Exemplar 1

Switch x and y

1 out of 1

Exemplar 2

flip the y -values

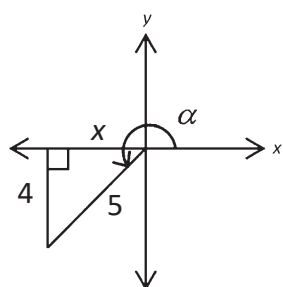
0 out of 1

Question 31

T6

Given that $\sin \alpha = -\frac{4}{5}$ where α is in quadrant III, and $\cos \beta = \frac{12}{13}$ where β is in quadrant IV, determine the exact value of $\cos(\alpha - \beta)$.

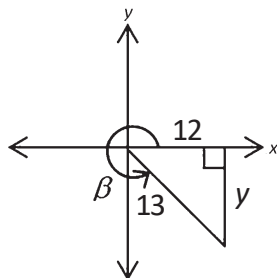
Solution



$$x^2 + 4^2 = 5^2$$

$$x^2 = 9$$

$$x = \pm 3$$



$$12^2 + y^2 = 13^2$$

$$y^2 = 25$$

$$y = \pm 5$$

½ mark for value of x

½ mark for value of y

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= \left(-\frac{3}{5}\right)\left(\frac{12}{13}\right) + \left(-\frac{4}{5}\right)\left(-\frac{5}{13}\right)$$

$$= \frac{-36 + 20}{65}$$

$$= -\frac{16}{65}$$

½ mark for consistent value of $\cos \alpha$

½ mark for consistent value of $\sin \beta$

1 mark for substitution into correct identity

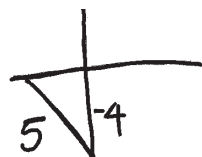
3 marks

Note:

Accept any of the following values for x : $x = \pm 3$; $x = -3$; or $x = 3$.

Accept any of the following values for y : $y = \pm 5$; $y = -5$; or $y = 5$.

Exemplar 1



For α

$$(5)^2 = (4)^2 + b^2$$

$$25 = 16 + b^2$$

$$9 = b^2$$

$$b = 3$$

$$\cos \alpha = -\frac{4}{5}$$

For β

$$(13)^2 = a^2 + (12)^2$$

$$169 = a^2 + 144$$

$$25 = a^2$$

$$a = 5$$

$$\sin \beta = -\frac{5}{13}$$

$$\begin{array}{r} 169 \\ -144 \\ \hline 25 \end{array}$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= \left(-\frac{4}{5}\right)\left(\frac{12}{13}\right) + \left(-\frac{3}{5}\right)\left(-\frac{5}{13}\right)$$

$$= -\frac{24}{65} + \frac{4}{13}$$

$$= \frac{-24 + 20}{65}$$

$$= -\frac{4}{65}$$

2½ out of 3

+ ½ mark for value of y

+ ½ mark for consistent value of $\cos \alpha$

+ ½ mark for consistent value of $\sin \beta$

+ 1 mark for substitution into correct identity

E3 (variables introduced without being defined)

Exemplar 2

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos\left(\frac{3}{5}\right) \cos\left(\frac{12}{13}\right) + \sin\left(-\frac{4}{5}\right)\left(\frac{5}{13}\right)$$

$$\cos(\alpha - \beta) = \left(\frac{3}{5}\right)\left(\frac{12}{13}\right) + \left(-\frac{4}{5}\right)\left(\frac{5}{13}\right)$$

$$\cos(\alpha - \beta) = \frac{39}{65} + \left(-\frac{20}{65}\right)$$

$$\boxed{\cos(\alpha - \beta) = \frac{19}{65}}$$

$$y^2 = r^2 - x^2$$

$$y^2 = 13^2 - 12^2$$

$$y^2 = 169 - 144$$

$$\sqrt{y^2} = \sqrt{25}$$

$$y = 5$$

$$x^2 = r^2 - y^2$$

$$x^2 = 5^2 - 4^2$$

$$x^2 = 25 - 16$$

$$\sqrt{x^2} = \sqrt{9}$$

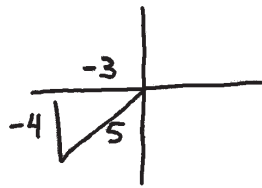
$$x = 3$$

1 out of 3

- + ½ mark for value of x
- + ½ mark for value of y
- + 1 mark for substitution into correct identity
- ½ mark for procedural error in line 2
- ½ mark for arithmetic error in line 4

Exemplar 3

$$\sin \alpha = -\frac{4}{5}$$

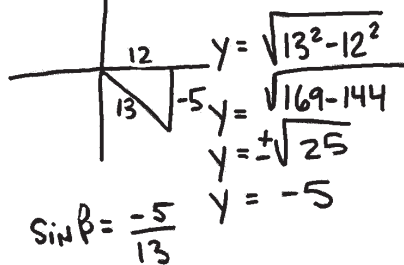


$$\begin{aligned}x &= \sqrt{5^2 - (-4)^2} \\x &= \sqrt{25 - 16} \\x &= \pm \sqrt{9} \\x &= -3\end{aligned}$$

$$\cos\left(\frac{-3}{5} - \frac{12}{13}\right) = \cos\left(\frac{-3}{5}\right)\cos\left(\frac{12}{13}\right) + \sin\left(\frac{-4}{5}\right)\left(\frac{-5}{13}\right)$$

$$\cos \alpha = \frac{-3}{5}$$

$$\cos \beta = \frac{12}{13}$$



$$\sin \beta = \frac{-5}{13}$$

2 out of 3

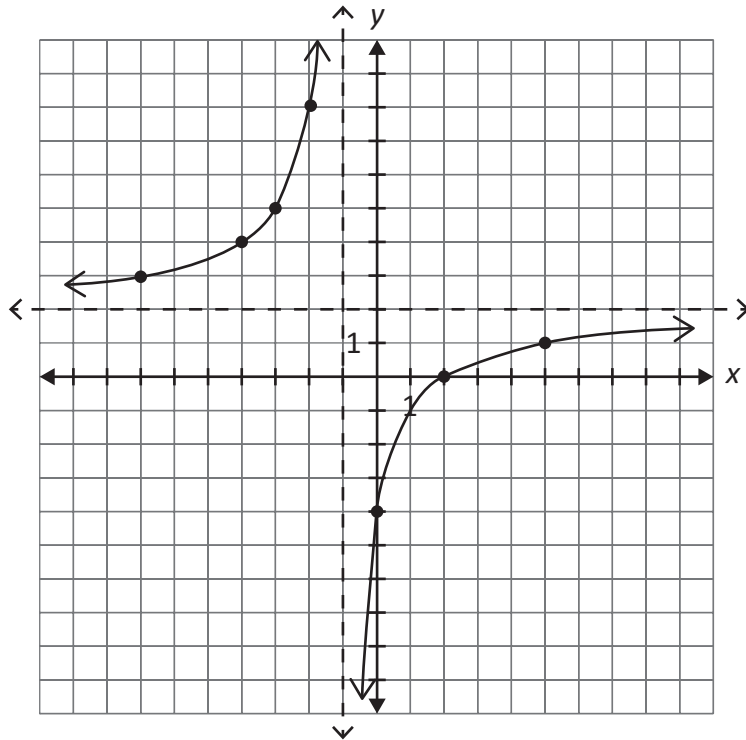
- + ½ mark for value of x
- + ½ mark for value of y
- + ½ mark for consistent value of $\cos \alpha$
- + ½ mark for consistent value of $\sin \beta$

Question 32

R14

Sketch the graph of $f(x) = \frac{2x-4}{x+1}$.

Solution



1 mark for asymptotic behaviour approaching $x = -1$

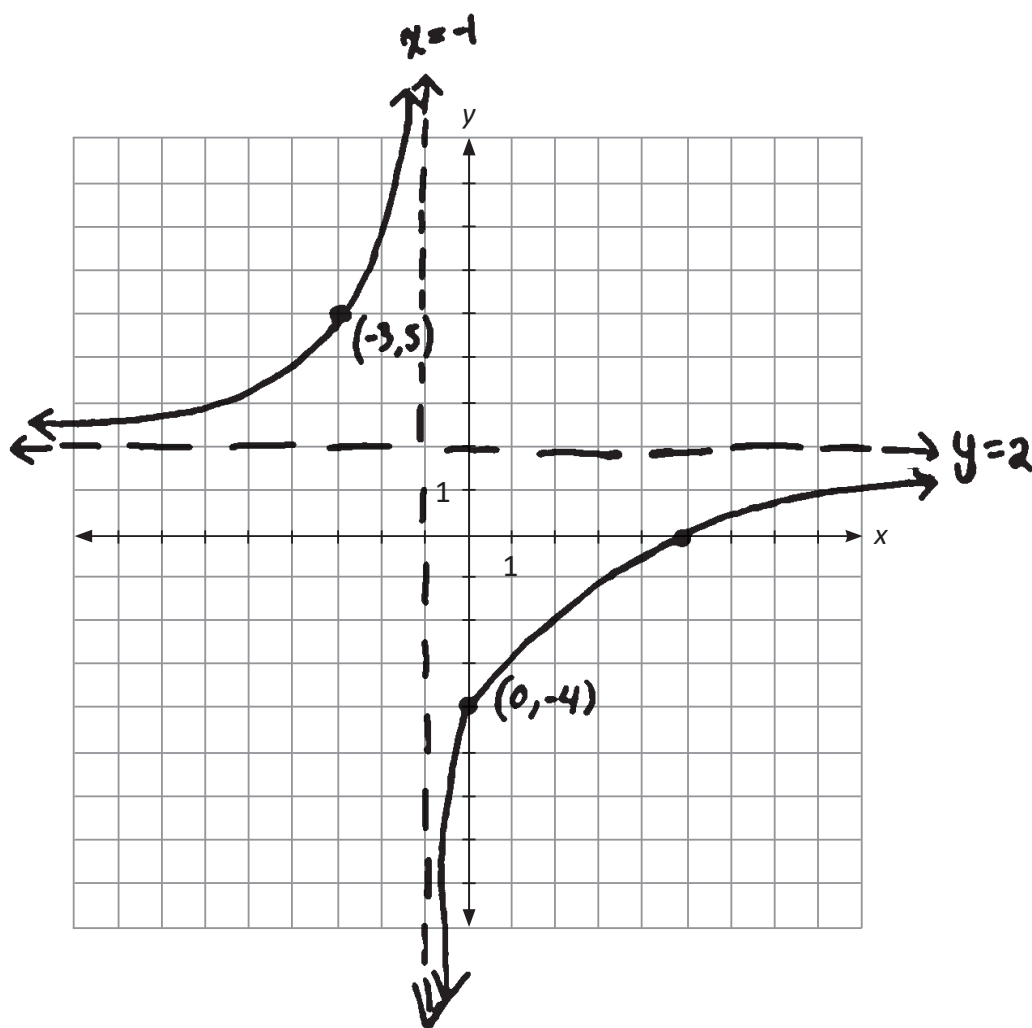
1 mark for asymptotic behaviour approaching $y = 2$

1 mark for shape of rational function

($\frac{1}{2}$ mark for graph left of $x = -1$, $\frac{1}{2}$ mark for graph right of $x = -1$)

3 marks

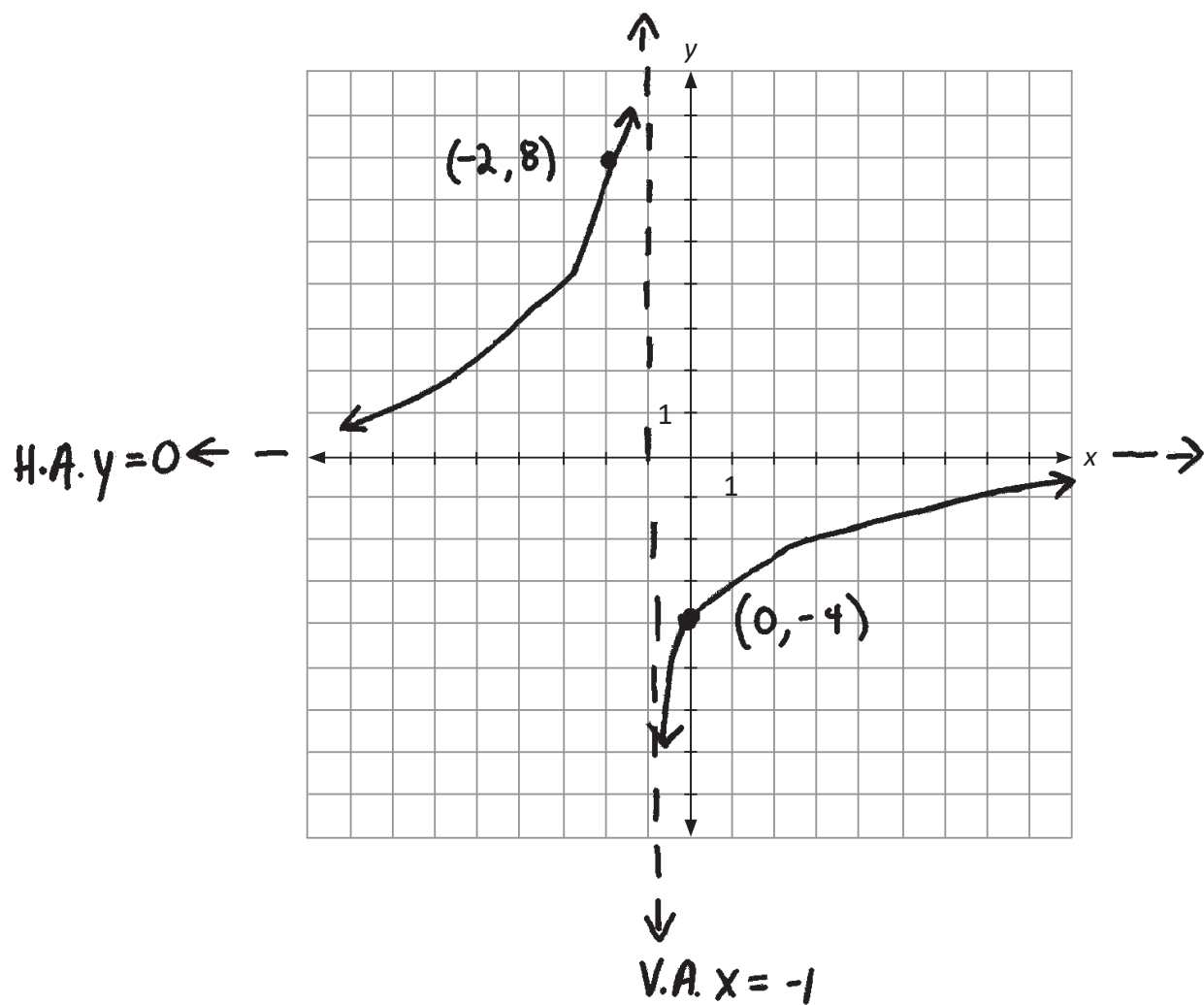
Exemplar 1



2½ out of 3

- + 1 mark for asymptotic behaviour approaching $x = -1$
- + 1 mark for asymptotic behaviour approaching $y = 2$
- + ½ mark for shape of graph left of $x = -1$

Exemplar 2

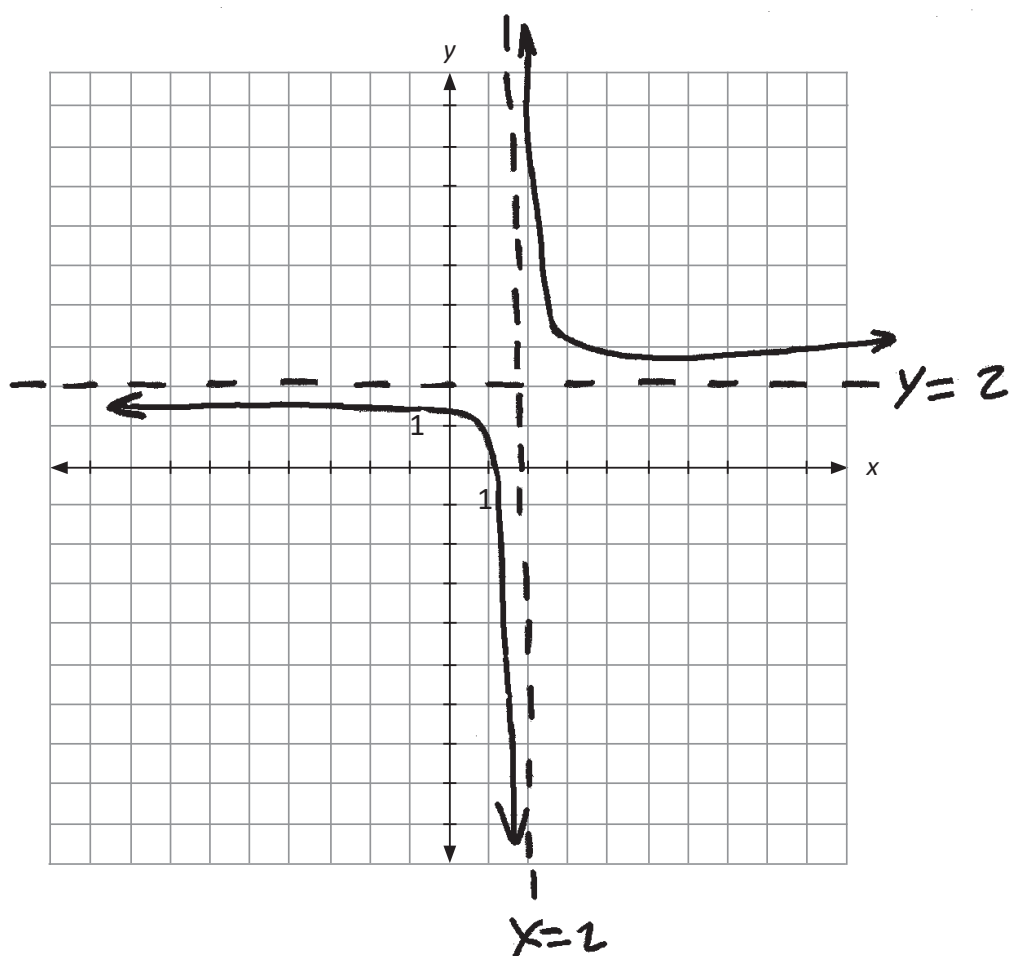


1½ out of 3

+ 1 mark for asymptotic behaviour approaching $x = -1$

+ ½ mark for shape of graph right of $x = -1$

Exemplar 3



1 out of 3

+ 1 mark for asymptotic behaviour approaching $y = 2$
E10 (graph crosses or curls away from asymptote)

Question 33

T3, T4

Due to the tidal variations, the depth of the water in a bay is represented by the function,

$$d(t) = -2\cos\left[\frac{\pi}{6}(t+4)\right] + 16$$

where d is the depth of the water, in metres and
 t is the time, in hours.

- a) Determine the depth of the water when $t = 2$.
- b) State the minimum depth of the water.
- c) Determine the length of time between two consecutive low tides (minimum depths).

Solution

$$\begin{aligned}\text{a) } d(2) &= -2\cos\left[\frac{\pi}{6}(2+4)\right] + 16 \\ &= -2\cos\pi + 16 \\ &= -2(-1) + 16 \\ &= 18 \text{ metres}\end{aligned}$$

1 mark for exact value

1 mark

b) 14 metres

1 mark

$$\begin{aligned}\text{c) period} &= \frac{2\pi}{\frac{\pi}{6}} \\ &= 12 \text{ hours}\end{aligned}$$

1 mark for period

1 mark

Exemplar 1

a)

$$\begin{aligned}d(a) &= -2 \cos\left[\frac{\pi}{6}(a+4)\right] + 16 \\&= -2 \cos(\pi) + 16 \\&= -2(1) + 16 \\&= 14 \text{ m}\end{aligned}$$

0 out of 1

b)

0 out of 1

c)

12

1 out of 1

award full marks

E5 (units of measure omitted in final answer)

Exemplar 2

a)

14 metres

0 out of 1

b)

10 metres

0 out of 1

c)

12 seconds

1 out of 1

award full marks

E5 (incorrect units of measure)

Question 34

T3

Evaluate.

$$\sin^2\left(\frac{5\pi}{4}\right)\sec\left(\frac{13\pi}{3}\right)-\cot\left(\frac{7\pi}{4}\right)$$

Solution

$$\left(-\frac{\sqrt{2}}{2}\right)^2\left(\frac{2}{1}\right)-(-1)$$

$$\frac{1+1}{2}$$

1 mark for $\sin\left(\frac{5\pi}{4}\right)$ (½ mark for value; ½ mark for quadrant)

1 mark for $\sec\left(\frac{13\pi}{3}\right)$ (½ mark for value; ½ mark for quadrant)

1 mark for $\cot\left(\frac{7\pi}{4}\right)$ (½ mark for value; ½ mark for quadrant)

3 marks

Exemplar 1

$$\begin{aligned} & \left(-\frac{\sqrt{2}}{2}\right)^2 (-2) (-1) \\ & \quad \cancel{\frac{2}{4}} \cancel{(-2)} \\ & \quad \underline{\underline{1}} \end{aligned}$$

2½ out of 3

+ 1 mark for $\sin\left(\frac{5\pi}{4}\right)$

+ ½ mark for value for $\sec\left(\frac{13\pi}{3}\right)$

+ 1 mark for $\cot\left(\frac{7\pi}{4}\right)$

E7 (transcription error in line 1)

Exemplar 2

$$\begin{aligned} &= \left(-\frac{\sqrt{2}}{2}\right)^2 \times \frac{1}{\frac{1}{2}} - \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \\ &= \frac{-2}{4} \times 2 - \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \\ &= -1 - \left(\frac{\sqrt{2}/2}{\sqrt{2}/2}\right) \\ &= -1 - 1 \\ &= -2 \end{aligned}$$

2 out of 3

+ 1 mark for $\sin\left(\frac{5\pi}{4}\right)$

+ 1 mark for $\sec\left(\frac{13\pi}{3}\right)$

+ ½ mark for value of $\cot\left(\frac{7\pi}{4}\right)$

– ½ mark for arithmetic error in line 2

Question 35

R14

a) State the non-permissible value(s) of the function, $f(x) = \frac{x-3}{x^2-9}$.

b) Describe what each non-permissible value represents on the graph of $f(x) = \frac{x-3}{x^2-9}$.

Solution

a) $x = 3, -3$

1 mark for non-permissible values ($\frac{1}{2}$ mark for each)

1 mark

b) At $x = 3$, there is a point of discontinuity (hole) on the graph of $f(x)$. At $x = -3$, there is a vertical asymptote on the graph of $f(x)$.

$\frac{1}{2}$ mark for point of discontinuity (hole) at $x = 3$

$\frac{1}{2}$ mark for vertical asymptote at $x = -3$

1 mark

Exemplar 1

a) $x \neq \pm 3$

1 out of 1

b) There is an asymptote at $x=-3$ and a point of discontinuity at $x=3$.

1 out of 1

Exemplar 2

a) $3, -3$

1 out of 1

b) $x=3$ is a vertical asymptote.
 $x=-3$ is a hole.

0 out of 1

Exemplar 3

a) $x = 3, -3$

1 out of 1

b) Represents the asymptotes.

½ out of 1

+ ½ mark for vertical asymptote at $x = -3$

Question 36

R7

Determine which of the given values is greater. Justify your answer.

$$\log_3 38 \text{ or } \log_4 50$$

Solution

$$3^3 = 27$$

$$3^4 = 81 \quad \log_3 38 > 3$$

$$4^2 = 16$$

$$4^3 = 64 \quad \log_4 50 < 3$$

$\therefore \log_3 38$ is greater

1 mark for justification

1 mark

Exemplar 1

$$\log_3 38 \text{ or } \log_4 50$$

↑
this one

0 out of 1

Exemplar 2

$\log_3 38$ is greater because

$$3^1 = 3$$

$$3^2 = 9$$

$$3^3 = 27$$

$$4^1 = 4$$

$$4^2 = 16$$

$$4^3 = 64$$

½ out of 1

award full marks

– ½ mark for lack of clarity

Exemplar 3

$$\log_3 38 = x$$

$$3^x = 38$$

$$3^3 = 27$$

$$3^4 = 81$$

$$\left. \begin{array}{l} 3^3 = 27 \\ 3^4 = 81 \end{array} \right\} \log_3 38 \approx 3.3$$

$$\log_4 50 = x$$

$$4^x = 50$$

$$4^2 = 16$$

$$4^3 = 64$$

$$\left. \begin{array}{l} 4^2 = 16 \\ 4^3 = 64 \end{array} \right\} \log_4 50 \approx 4.7$$

$\therefore \log_4 50$ is the greater value.

½ out of 1

award full marks

– ½ mark for procedural error

Determine if the point $\left(\frac{3}{4}, \frac{\sqrt{5}}{4}\right)$ is on the unit circle. Justify your answer.

Solution

$$x^2 + y^2 = 1$$

$$\text{LHS} = \left(\frac{3}{4}\right)^2 + \left(\frac{\sqrt{5}}{4}\right)^2$$

$$= \frac{9}{16} + \frac{5}{16}$$

$$= \frac{14}{16}$$

$$\frac{14}{16} \neq 1$$

$\therefore$ The point is not on the unit circle.

1 mark for justification

1 mark

Exemplar 1

$$x^2 + y^2 = 1$$

$$\left(\frac{3}{4}\right)^2 + \left(\frac{\sqrt{5}}{4}\right)^2 = 1$$

$$\frac{9}{16} + \frac{5}{16} = 1$$

$$\frac{14}{16} \neq 1$$

$\frac{14}{16}$ does not equal 1 so this point

is not on the unit circle.

1 out of 1

award full marks

E2 (equating the two sides when proving an identity)

Exemplar 2

$$x^2 + y^2 = r^2$$

$$\frac{3^2}{4} + \frac{\sqrt{5}^2}{4}$$

$$\frac{9}{16} + \frac{5}{16}$$

$$\frac{14}{16} \neq 1$$

no it is not

1 out of 1

award full marks

E4 (missing brackets but still implied in line 2)

Exemplar 3

$$\left(\frac{3}{4}\right)^2 + \left(\frac{\sqrt{5}}{4}\right)^2$$

$$\frac{9}{16} + \frac{\sqrt{25}}{16}$$

$$\frac{9}{16} + \frac{5}{16} = \frac{16}{16}$$

$$= 1$$

yes because it equals 1

½ out of 1

award full marks

– ½ mark for arithmetic error in line 3

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Question 38

P4

Determine which term contains x^5 in the binomial expansion of $\left(2x + \frac{3}{x}\right)^{11}$.

Solution

Method 1

$$x^{11}, (x^{10})\left(\frac{1}{x}\right), (x^9)\left(\frac{1}{x^2}\right), \dots$$

1 mark for determining a pattern

$$x^{11}, x^9, x^7, \dots$$

$\therefore$ The fourth term contains x^5 .

1 mark for fourth term
(or a term consistent with the pattern)

2 marks

Method 2

$$(x)^{11-k} (x^{-1})^k = x^5$$

$$x^{11-2k} = x^5$$

$$11 - 2k = 5$$

$$6 = 2k$$

$$k = 3$$

$\frac{1}{2}$ mark for substitution

$\frac{1}{2}$ mark for solving for k

$\therefore$ The fourth term contains x^5 .

1 mark for fourth term
(or a term consistent with the value of k)

2 marks

Exemplar 1

$$T_1 \rightarrow {}^{11}C_0 (2x)^{11} \left(\frac{3}{x}\right)^0$$

$$(0) (2x^{11}) (0)$$

$$=? x^{11}$$

$$T_2 \rightarrow {}^{11}C_1 (2x)^{10} \left(\frac{3}{x}\right)^1$$

$$(1) (x^{10}) \left(\frac{3}{x^1}\right)$$

$$x^9$$

$$\frac{x^{10}}{x^1}$$

$$11 \text{ to } 9 = -2$$

$$\begin{array}{ccc} 9, 7, 5 \\ \uparrow \uparrow \uparrow \\ T_2 T_3 T_4 \end{array}$$

term 4 will
contain x^5

1½ out of 2

award full marks

– ½ mark for procedural errors in lines 2 and 5

Exemplar 2

$$t_{k+1} = {}^nC_k a^{n-k} b^k$$

$$= {}^{11}C_k (2x)^{11-k} \left(\frac{3}{x}\right)^k \rightarrow x^5$$

$$x^{11-k} = x^5$$

$$11-k = 5$$

$$11-5 = k$$

$$6 = k$$

→ the 7th term contains x^5

1½ out of 2

+ ½ mark for substitution

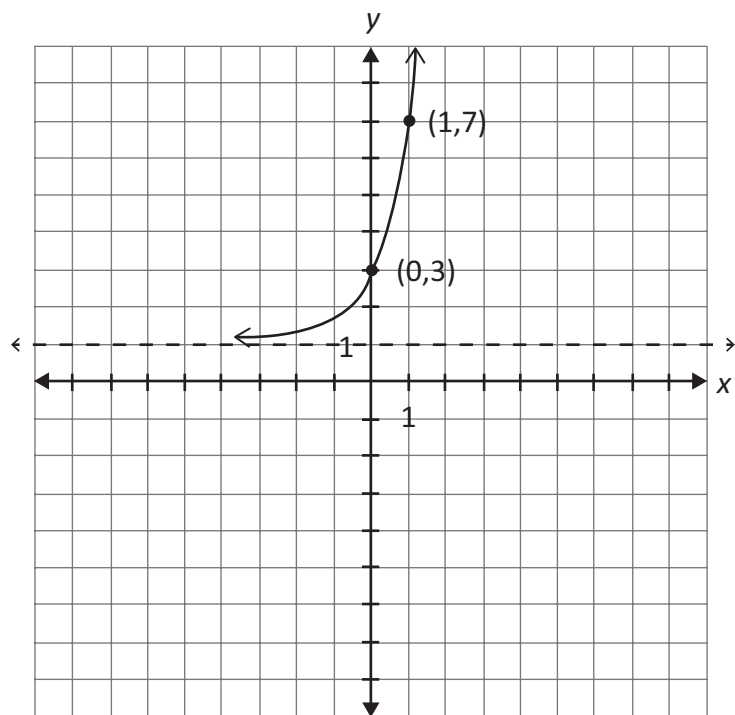
+ 1 mark for term consistent with the value of k

Question 39

R9

Sketch the graph of $y = 2(3^x) + 1$.

Solution



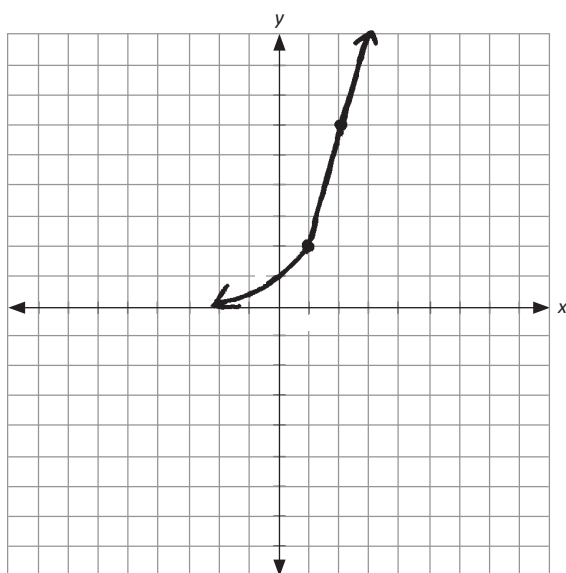
1 mark for shape of an increasing exponential function

1 mark for vertical stretch

1 mark for vertical translation

3 marks

Exemplar 1



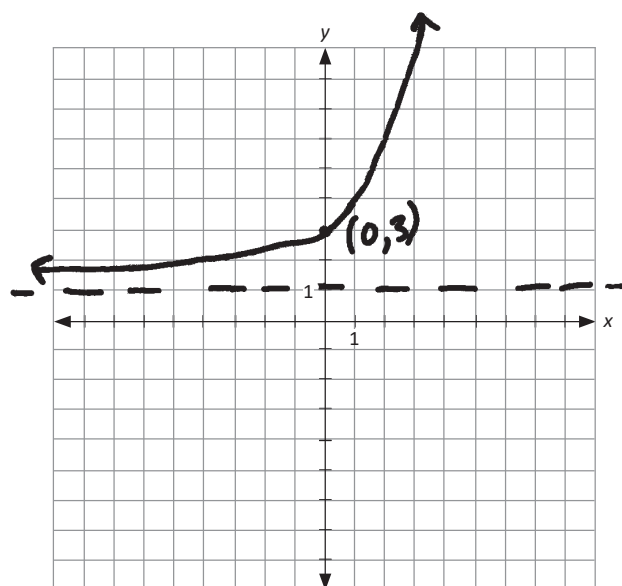
2 out of 3

+ 1 mark for shape of an increasing exponential function

+ 1 mark for vertical stretch

E10 (asymptote omitted but still implied)

Exemplar 2

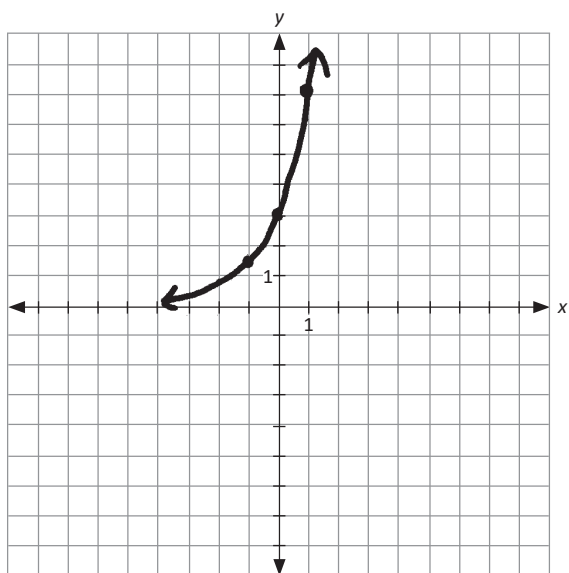


2½ out of 3

award full marks

– ½ mark for procedural error (not showing second point on the graph)

Exemplar 3



2 out of 3

+ 1 mark for shape of an increasing exponential function

+ 1 mark for vertical stretch

E10 (asymptote omitted but still implied)

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Question 40

R8, R10

Solve, algebraically.

$$\log(2x+1) = \log(x-2) + 1$$

Solution

Method 1

$$\log(2x+1) - \log(x-2) = 1$$

$$\log\left(\frac{2x+1}{x-2}\right) = 1$$

$$10^1 = \frac{2x+1}{x-2}$$

$$10x - 20 = 2x + 1$$

$$8x = 21$$

$$x = \frac{21}{8}$$

1 mark for quotient law

1 mark for exponential form

2 marks

Method 2

$$\log(2x+1) = \log(x-2) + \log 10$$

$$\log(2x+1) = \log(10(x-2))$$

$$2x+1 = 10(x-2)$$

$$2x+1 = 10x-20$$

$$21 = 8x$$

$$x = \frac{21}{8}$$

½ mark for logarithmic form

1 mark for product law

½ mark for equating arguments

2 marks

Exemplar 1

$$\log(2x+1) - \log(x-2) = 1$$

$$\log\left(\frac{2x+1}{x-2}\right) = 1$$

$$\frac{2(x + \frac{1}{2})}{x-2}$$

1 out of 2

+ 1 mark for quotient law

Exemplar 2

$$\log(2x+1) - \log(x-2) = 1$$

$$\log\left(\frac{2x+1}{x-2}\right) = 1$$

$$\frac{2x+1}{x-2} = 10(x-2)$$

$$\begin{array}{r} 2x+1 = 10x-20 \\ -2x \quad -2x+20 \\ \hline +20 \end{array}$$

$$\frac{19}{8} = \frac{8x}{8}$$

$$\boxed{\frac{9}{8} = x}$$

2 out of 2

award full marks

E7 (transcription error in lines 4 and 6)

E1 (impossible solution not rejected in final answer)

Exemplar 3

$$\log(2x+1) - \log(x-2) = 1$$

$$\frac{2x+1}{x-2} = 1$$

$$2x+1 = x-2$$

$$x = -3$$

0 out of 2

+ 1 mark for quotient law

– 1 mark for concept error (omitting log)

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Question 41

R1

Given the functions, $f(x) = x + 2$ and $g(x) = x - 2$,

a) state the equation of $h(x) = f(x) \cdot g(x)$.

b) sketch the graph of $h(x) = f(x) \cdot g(x)$.

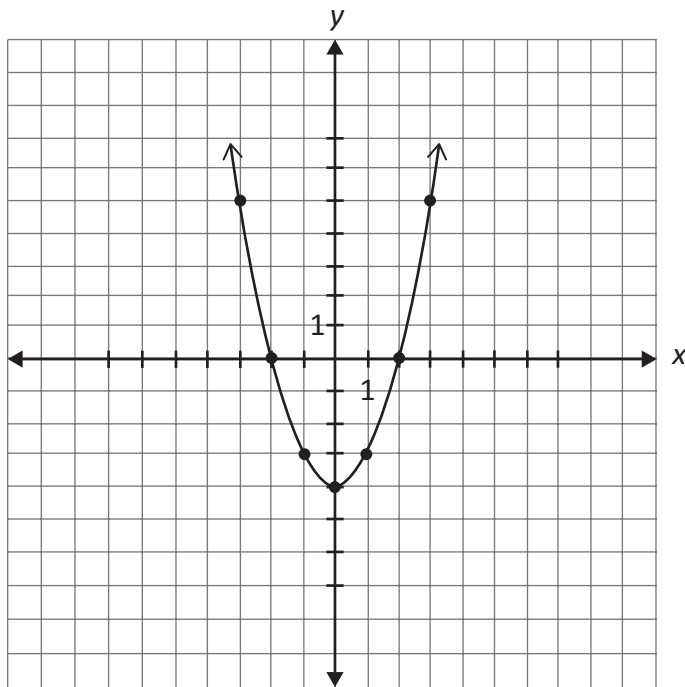
Solution

a) $h(x) = (x + 2)(x - 2)$

or

$$h(x) = x^2 - 4$$

b)



1 mark

1 mark for graph consistent with a)

1 mark

Exemplar 1

a)

$$h(x) = (x+2)(x-2)$$

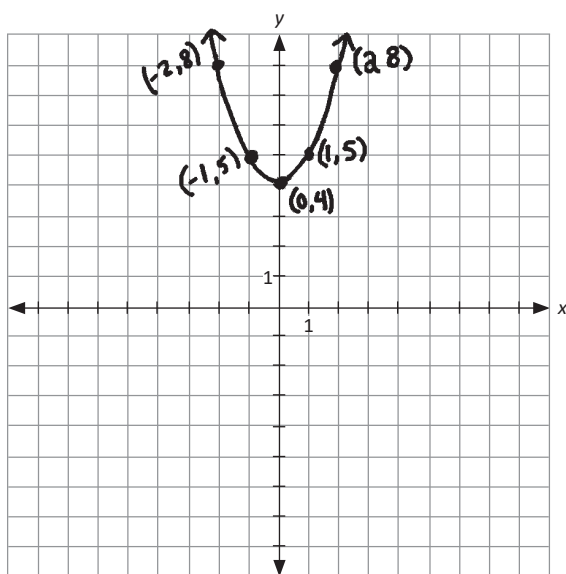
$$h(x) = x^2 + 2x - 2x + 4$$

½ out of 1

award full marks

– ½ mark for arithmetic error in line 2

b)



1 out of 1

Exemplar 2

a)

$$(x+2)(x-2)$$

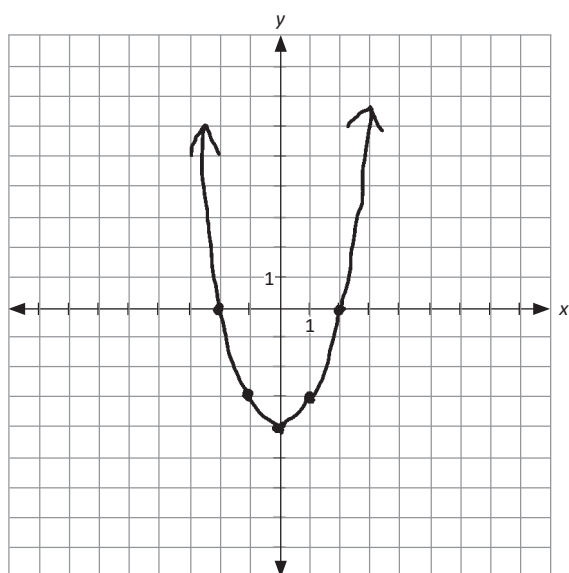
$$(x^2 - 4)$$

1 out of 1

award full marks

E2 (changing an equation to an expression)

b)



1 out of 1

Exemplar 3

a)

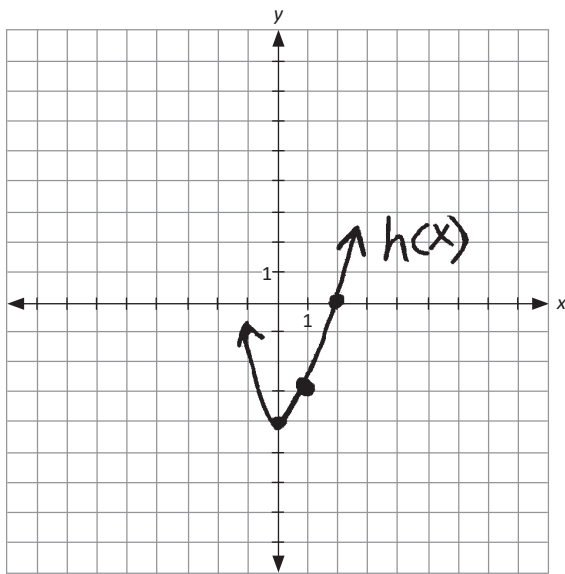
$$h(x) = (x+2)(x-2)$$

$$h(x) = x^2 - 2x + 2x - 4$$

$$h(x) = x^2 - 4$$

1 out of 1

b)



½ out of 1

award full marks

– ½ mark for procedural error (one incorrect point)

Exemplar 4

a)

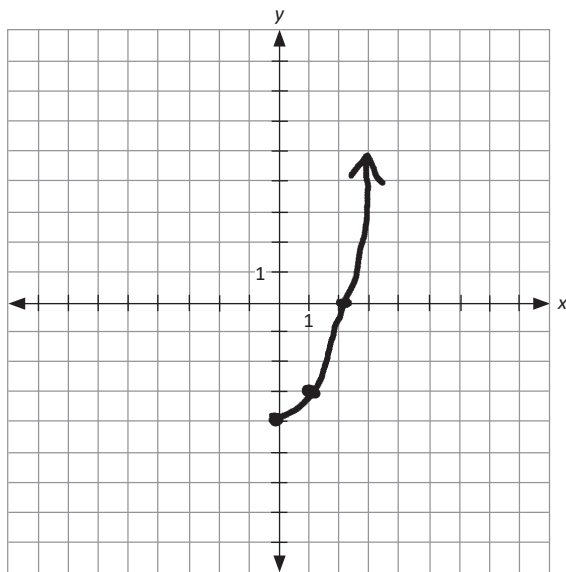
$$h(x) = (x+2)(x-2)$$
$$x^2 - \cancel{2x} + \cancel{2x} - 4$$
$$x^2 - 4$$

1 out of 1

award full marks

E2 (changing an equation to an expression)

b)



½ out of 1

award full marks

– ½ mark for procedural error (restricting domain)

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Question 42

R11

Given $p(x) = -2x^3 + 3x + 1$, determine the remainder when $p(x)$ is divided by $(x - 2)$.

Solution

Method 1

$$p(2) = -2(2)^3 + 3(2) + 1$$

$$p(2) = -2(8) + 6 + 1$$

$$p(2) = -9$$

1 mark for remainder theorem

1 mark

Method 2

2		-2	0	3	1
		↓	-4	-8	-10
		-2	-4	-5	-9

1 mark for synthetic division (or any equivalent strategy)

1 mark

∴ The remainder is -9

Exemplar 1

$$\begin{array}{r|rrrr} 2 & -2 & 0 & 3 & 1 \\ & \downarrow & -4 & -8 & -10 \\ \hline & -2 & -4 & -5 & 9 \end{array}$$

$$\text{remainder} = 9$$

½ out of 1

award full marks

– ½ mark for arithmetic error

Exemplar 2

$$\begin{array}{r|rrr} 2 & -2 & 3 & 1 \\ & & -4 & -2 \\ \hline + & & & \\ \hline x & -2 & -1 & (-1) \end{array}$$

½ out of 1

award full marks

– ½ mark for procedural error (missing coefficient of zero)

Exemplar 3

$$\begin{array}{r|rrrr} 2 & 2 & 0 & 3 & 1 \\ & \downarrow & 4 & 8 & 22 \\ \hline & 2 & 4 & 11 & 23 \end{array}$$

$$2x^2 + 4x + 11 + \frac{23}{x-2}$$

1 out of 1

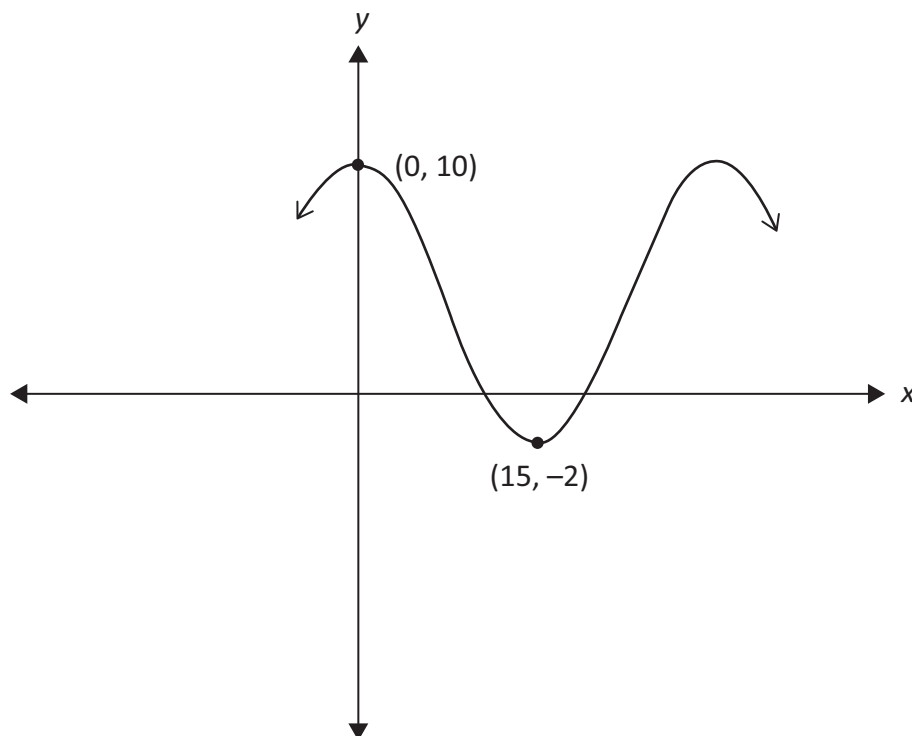
award full marks

E7 (transcription error)

E1 (final answer not stated)

Question 43**T4**

Determine the equation of the sinusoidal function in the form $y = A\cos(Bx) + D$.

**Solution**

$$y = 6\cos\left(\frac{\pi}{15}x\right) + 4$$

1 mark for the value of A

1 mark for the value of B ($\frac{1}{2}$ mark for period; $\frac{1}{2}$ mark for consistent value of B)

1 mark for the value of D

3 marks

Exemplar 1

$$y = 6 \cos\left(\frac{\pi}{15}\right) + 4$$

3 out of 3

award full marks

E3 (variable omitted in an equation)

Exemplar 2

$$a = 6$$
$$b = \frac{2\pi^{15}}{2\pi}$$
$$y = 6 \cos\left(\frac{15}{\pi} x\right) - 2$$

1½ out of 3

+ 1 mark for the value of A

+ ½ mark for period

Question 44

P2

Determine the total number of arrangements of the letters in the word TORONTO.

State your answer as a whole number.

Solution

$$\frac{7!}{2!3!}$$
$$\frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot \cancel{3!}}{2! \cancel{3!}}$$
$$\frac{7 \cdot 6 \cdot 5 \cdot 4}{2}$$
$$7 \cdot 6 \cdot 5 \cdot 2$$
$$420$$

½ mark for 7!

½ mark for division by 2!3!

½ mark for factorial expansion

½ mark for consistent number of arrangements

2 marks

Exemplar 1

$$\frac{7!}{2! 3!} \quad \begin{array}{r} 42.5 - 4 \\ 7 \cdot 6 \cdot 5 \cdot 4 \cdot \cancel{3} \cdot \cancel{2} \cdot 1 \\ \hline \cancel{2} \cdot \cancel{2} \end{array}$$

1 out of 2

+ ½ mark for 7!

+ ½ mark for division by 2!3!

Exemplar 2

$$\frac{7!}{2! 3!}$$

1 out of 2

+ ½ mark for 7!

+ ½ mark for division by 2!3!

Exemplar 3

$$\begin{array}{r} 7! \\ 3! 2! \end{array} \quad \begin{array}{l} b=3 \\ T=2 \end{array} \quad \text{total}=7$$
$$\Rightarrow \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot \cancel{3}}{2! (\cancel{3}!)}$$

$$\frac{7 \cdot 6 \cdot 5 \cdot 4}{2}$$

1½ out of 2

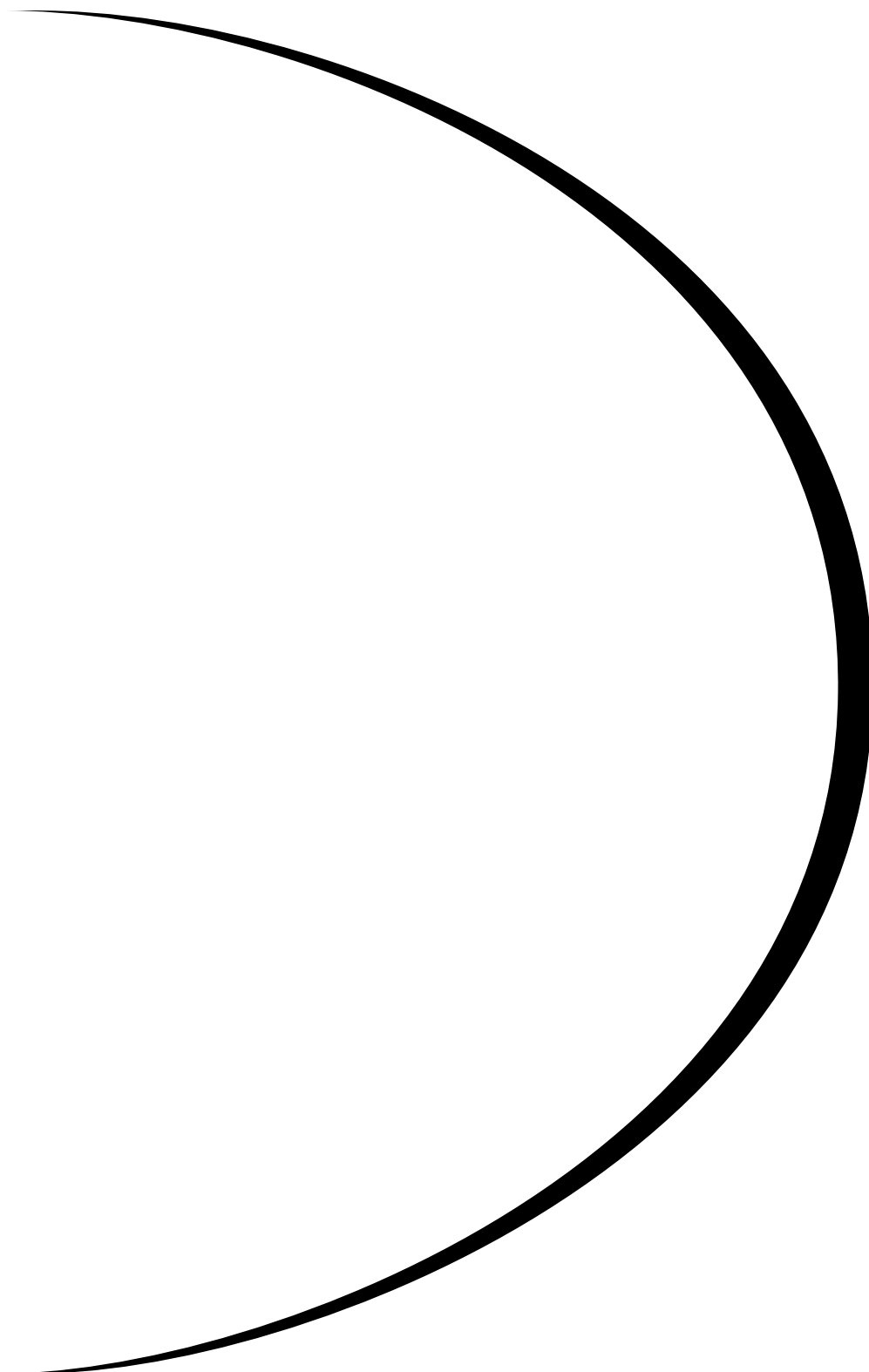
+ ½ mark for 7!

+ ½ mark for division by 2!3!

+ ½ mark for factorial expansion

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Appendices



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Appendix A

Marking Guidelines

Errors that are conceptually related to the learning outcomes associated with the question will result in a 1 mark deduction.

Each time a student makes one of the following errors, a $\frac{1}{2}$ mark deduction will apply:

- arithmetic error
- procedural error
- terminology error in explanation
- lack of clarity in explanation, description, or justification
- incorrect shape of graph (only when marks are not allocated for shape)

Communication Errors

The following errors, which are not conceptually related to the learning outcomes associated with the question, may result in a $\frac{1}{2}$ mark deduction and will be tracked on the *Answer/Scoring Sheet*.

E1 final answer	▪ answer given as a complex fraction ▪ final answer not stated ▪ impossible solution(s) not rejected in final answer and/or in step leading to final answer
E2 equation/expression	▪ changing an equation to an expression or vice versa ▪ equating the two sides when proving an identity
E3 variables	▪ variable omitted in an equation or identity ▪ variables introduced without being defined
E4 brackets	▪ "sin x²" written instead of "sin² x" ▪ missing brackets but still implied
E5 units	▪ units of measure omitted in final answer ▪ incorrect units of measure ▪ answer stated in degrees instead of radians or vice versa
E6 rounding	▪ rounding error ▪ rounding too early
E7 notation/transcription	▪ notation error ▪ transcription error
E8 domain/range	▪ answer outside the given domain ▪ bracket error made when stating domain or range ▪ domain or range written in incorrect order
E9 graphing	▪ endpoints or arrowheads omitted or incorrect ▪ scale values on axes not indicated or incorrectly spaced ▪ coordinate points labelled incorrectly
E10 asymptotes	▪ asymptotes drawn as solid lines ▪ asymptotes omitted but still implied ▪ graph crosses or curls away from asymptotes

Appendix B

Irregularities in Provincial Tests

A Guide for Local Marking

During the marking of provincial tests, irregularities are occasionally encountered in test booklets. The following list provides examples of irregularities for which an *Irregular Test Booklet Report* should be completed and sent to the department:

- completely different penmanship in the same test booklet
- incoherent work with correct answers
- notes from a teacher indicating how he or she has assisted a student during test administration
- student offering that he or she received assistance on a question from a teacher
- student submitting work on unauthorized paper
- evidence of cheating or plagiarism
- disturbing or offensive content
- no responses provided by the student or only incorrect responses (“0”)

Student comments or responses indicating that the student may be at personal risk of being harmed or of harming others are personal safety issues. This type of student response requires an immediate and appropriate follow-up at the school level. In this case, please ensure the department is made aware that follow-up has taken place by completing an *Irregular Test Booklet Report*.

Except in the case of cheating or plagiarism where the result is a provincial test mark of 0%, it is the responsibility of the division or the school to determine how they will proceed with irregularities. Once an irregularity has been confirmed, the marker prepares an *Irregular Test Booklet Report* documenting the situation, the people contacted and the follow-up. The original copy of this report is to be returned by the local jurisdiction and a copy is to be sent to the department along with the test materials.

Irregular Test Booklet Report

Test: _____

Date marked: _____

Booklet No.: _____

Problem(s) noted: _____

Question(s) affected: _____

Action taken or rationale for assigning marks: _____

Follow up: _____

Decision: _____

Marker's Signature: _____

Principal's Signature: _____

For Department Use Only - After Marking Complete

Consultant: _____

Date: _____

Appendix C

Table of Questions by Unit and Learning Outcome

Unit A: Transformations of Functions		
Question	Learning Outcome	Mark
7	R3, R5	3
11	R1	3
14	R1	2
17	R2	1
25	R1	1
30	R6	1
41a)	R1	1
41b)	R1	1
Unit B: Trigonometric Functions		
Question	Learning Outcome	Mark
2	T1	2
18	T3	1
20	T3	1
27	T3	2
33a)	T3	1
33b)	T4	1
33c)	T4	1
34	T3	3
37	T2	1
43	T4	3
Unit C: Binomial Theorem		
Question	Learning Outcome	Mark
1	P1	1
3	P3	3
6	P1	2
21	P4	1
38	P4	2
44	P2	2
Unit D: Polynomial Functions		
Question	Learning Outcome	Mark
9	R11	1
15	R12	3
23	R12	1
26	R11	2
42	R11	1

Unit E: Trigonometric Equations and Identities		
Question	Learning Outcome	Mark
5	T5	4
12a)	T6	1
12b)	T6	3
22	T5	1
24	T6	1
31	T6	3
Unit F: Exponents and Logarithms		
Question	Learning Outcome	Mark
4	R10	3
10	R8	3
19	R7	1
29	R8	2
36	R7	1
39	R9	3
40	R8, R10	2
Unit G: Radicals and Rationals		
Question	Learning Outcome	Mark
8	R13	2
13	R13	2
16	R14	2
28	R13	3
32	R14	3
35a)	R14	1
35b)	R14	1